Configuration and provenance

Standalone EJ-230 report
Branch: `feat/ej230-sslg4`
Generation source commit: `0b8be529e641`
Data session: `session_20260612_171407` on `t0minidaq`
Data path: `results_ej230/data`; generated 2026-06-12
Report generation: 2026-06-13 11:16 CEST

- 31 positions × 2000 events; main scan produced on `t0minidaq` with 24 threads.
- Directed controls on `MSI` with 16 threads: 4 EndTop × 2000 events and 3 End-only × 10000 events.
- 16 End + 70 Top channels; jitter fixed to zero.
- EJ-230 via SSLG4 `OPSC-106`: yield 9700/MeV, $\tau_r = 0.5$ ns, $\tau_d = 1.5$ ns, ABSLENGTH=120 cm.

Scope

Top 70 SiPM is a simulation coverage study. Real hardware: 32 Top SiPMs. All reported timing is intrinsic, without SPTR or electronics jitter.

EndTop geometry and channel map

- EJ-230 bar: $1400 \times 60 \times 10$ mm^3.
- End L IDs 0–7; End R IDs 8–15.
- Top L IDs 16–50: $-692, -672, \ldots, -12$ mm.
- Top R IDs 51–85: $+12, +32, \ldots, +692$ mm.
- Each Top element and window: 6×6 mm^2.

Exact constraints

First centers are 5 mm from each bar end. Pitch is 20 mm except the deliberate central $-12/+12$ mm pair, whose pitch is 24 mm.

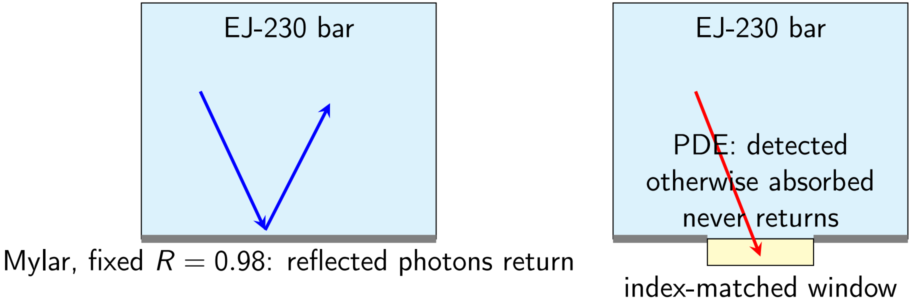
Instrumented and wrapped faces

Face	Optical state	Readout
$-X$, $+X$	open	8 End SiPMs each
$+Y$	Mylar with 70 exact windows	70 Top SiPMs
$-Y$, $-Z$, $+Z$	complete Mylar	none

External layer

Black Tedlar is deliberately not modeled; it only blocks ambient light. The non-reflected Mylar fraction is already absorbed in the optical model.

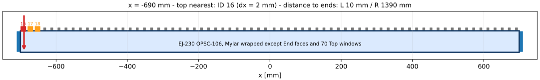
Window optics: why timing and collection change

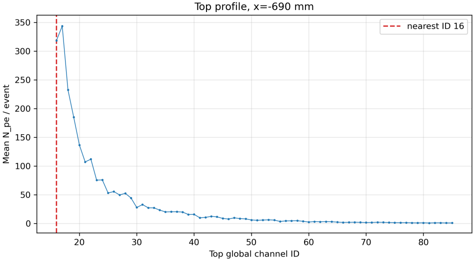


Consequence

Every window is an optical drain. Long, multiply reflected paths have more chances to meet one, so late End photons are preferentially removed.

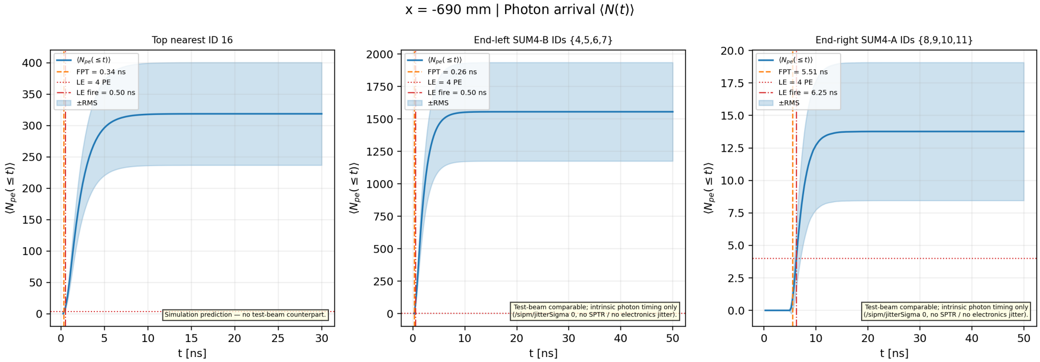
Position x=-690 mm



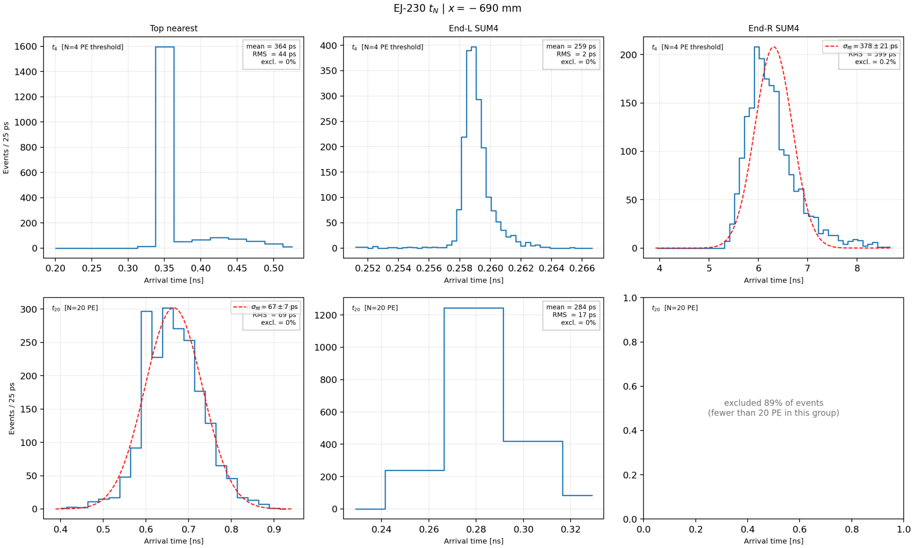


Quantity	Value
End L N_{pe}	3073.3 ± 16.8
End R N_{pe}	27.4 ± 0.2
Nearest Top ID	16 (window-track exception)
Nearest Top N_{pe}	318.7 ± 1.8
Nearest Top $\langle t \rangle$	2.3460 ± 0.0020 ns
Nearest Top FPT	0.34237 ± 0.00012 ns

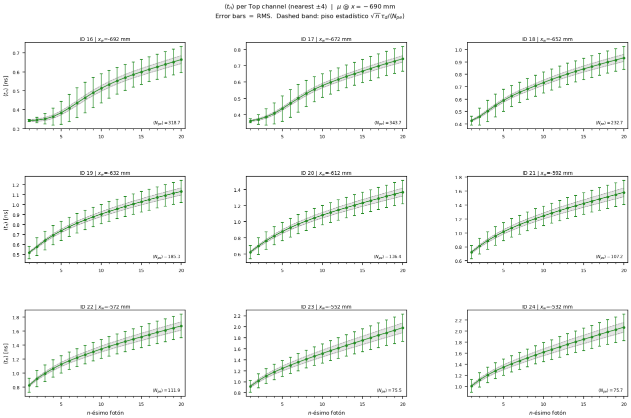
Run 1 | x = -690 mm | Photon arrival $\langle N(t) \rangle$

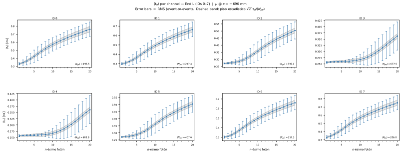


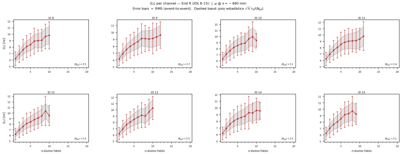
Run 1 | x = -690 mm | Time-to-threshold t_N



Run 1 | x = -690 mm | $\langle t_n \rangle$ per channel (dispersion)

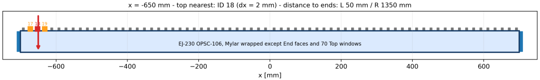


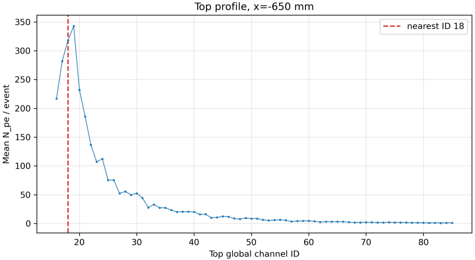




Error bars = RMS (event-to-event). Dashed band: statistical floor $\sigma_{stat}(t_n) \approx \sqrt{n}\,\tau_d/\langle N_{pe}\rangle$ ($\tau_d = 1.5$ ns). See dispersion-decomposition frame for physical interpretation.

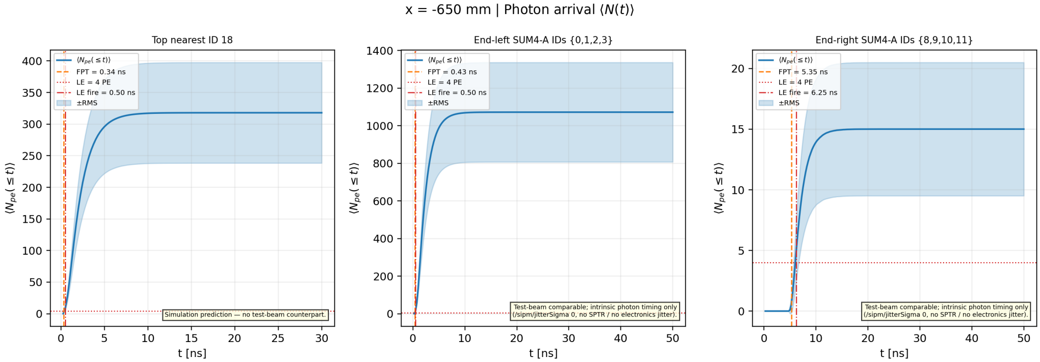
Position x=-650 mm



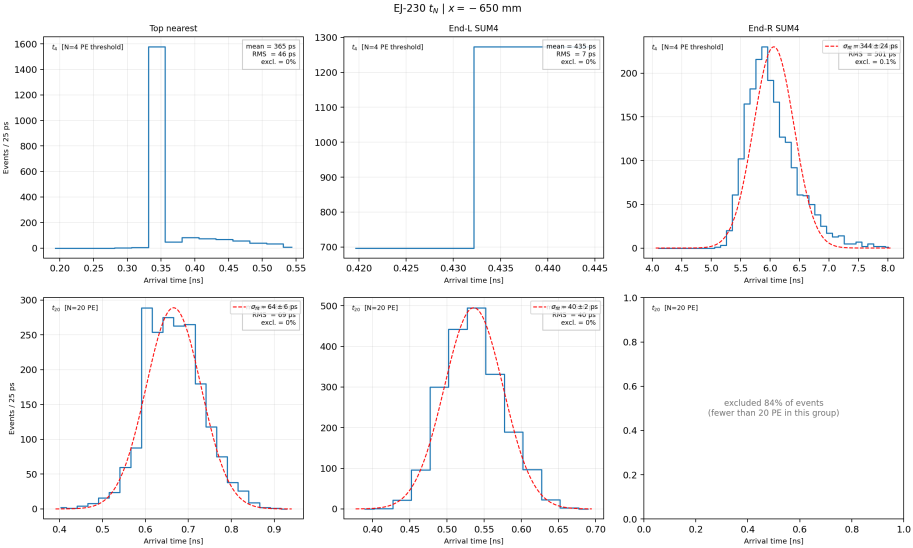


Quantity	Value
End L N_{pe}	2124.3 ± 11.6
End R N_{pe}	29.9 ± 0.2
Nearest Top ID	18 (window-track exception)
Nearest Top N_{pe}	318.0 ± 1.8
Nearest Top $\langle t \rangle$	2.3394 ± 0.0020 ns
Nearest Top FPT	0.34248 ± 0.00020 ns

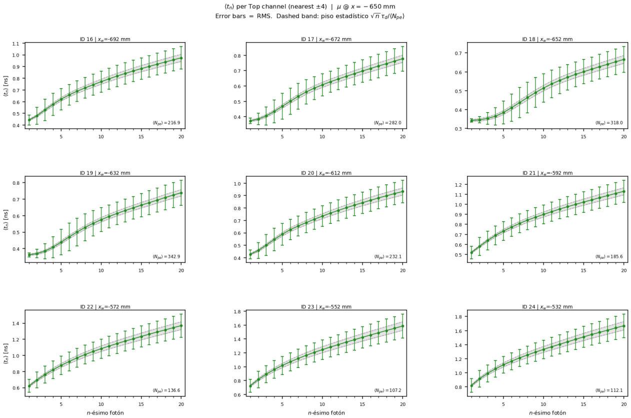
Run 2 | x = -650 mm | Photon arrival $\langle N(t) \rangle$

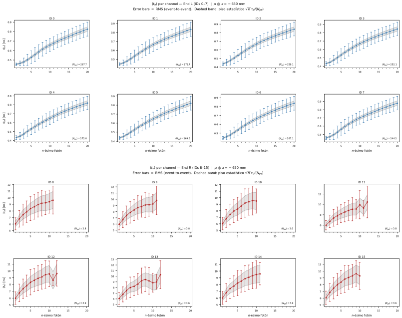


Run 2 | x = -650 mm | Time-to-threshold t_N



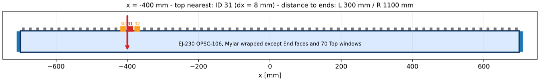
Run 2 | x = -650 mm | $\langle t_n \rangle$ per channel (dispersion)

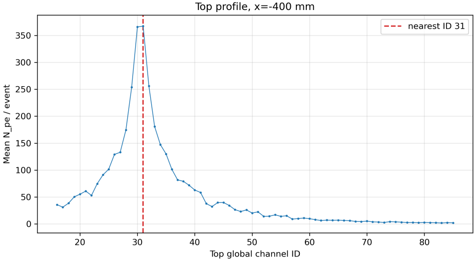




Error bars = RMS (event-to-event). Dashed band: statistical floor $\sigma_{stat}(t_n) \approx \sqrt{n}\,\tau_d/\langle N_{pe}\rangle$ ($\tau_d = 1.5$ ns). See dispersion-decomposition frame for physical interpretation.

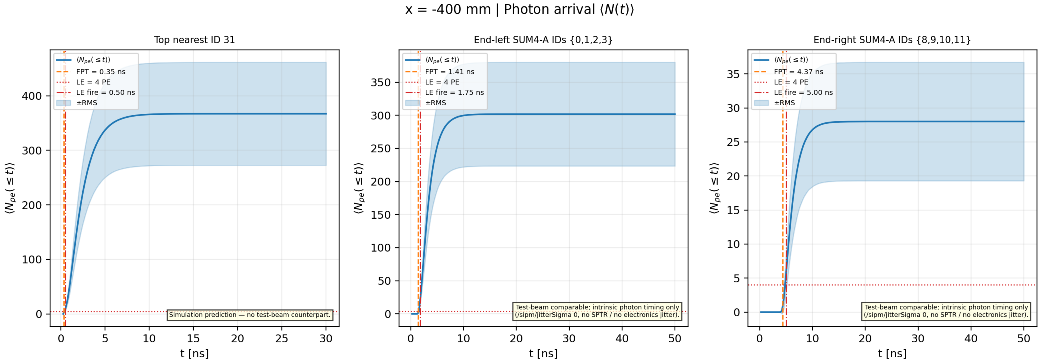
Position x=-400 mm



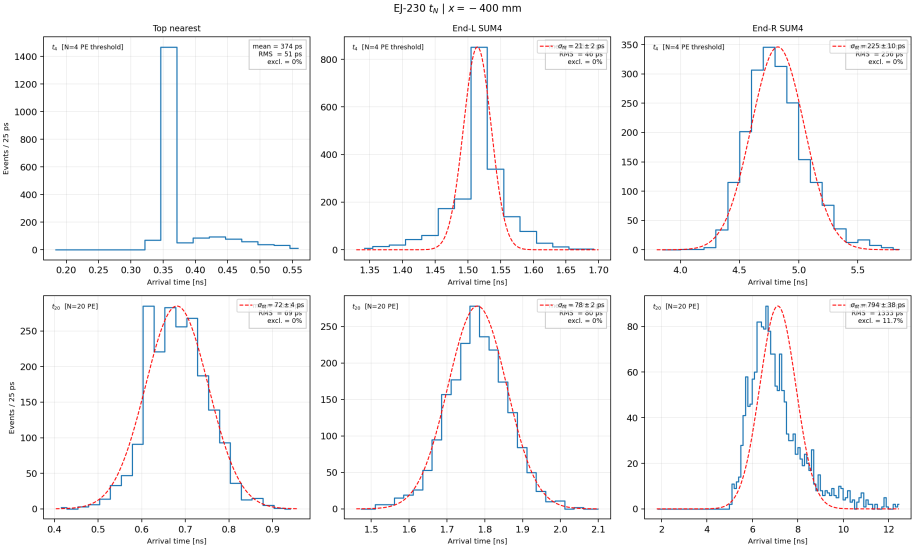


Quantity	Value
End L N_{pe}	581.1 ± 3.4
End R N_{pe}	53.0 ± 0.3
Nearest Top ID	31 (exact geometry)
Nearest Top N_{pe}	367.2 ± 2.1
Nearest Top $\langle t \rangle$	2.4926 ± 0.0020 ns
Nearest Top FPT	0.34578 ± 0.00019 ns

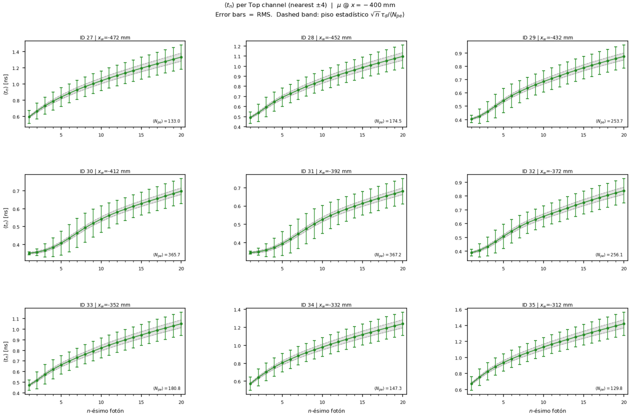
Run 3 | x = -400 mm | Photon arrival $\langle N(t)\rangle$

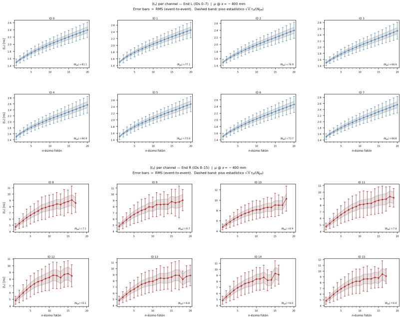


Run 3 | x = -400 mm | Time-to-threshold t_N



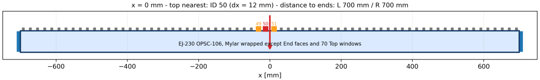
Run 3 | x = -400 mm | $\langle t_n \rangle$ per channel (dispersion)

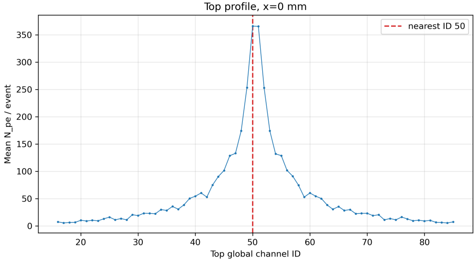




Error bars = RMS (event-to-event). Dashed band: statistical floor $\sigma_{stat}(t_n) \approx \sqrt{n}\,\tau_d/\langle N_{pe}\rangle$ ($\tau_d = 1.5$ ns). See dispersion-decomposition frame for physical interpretation.

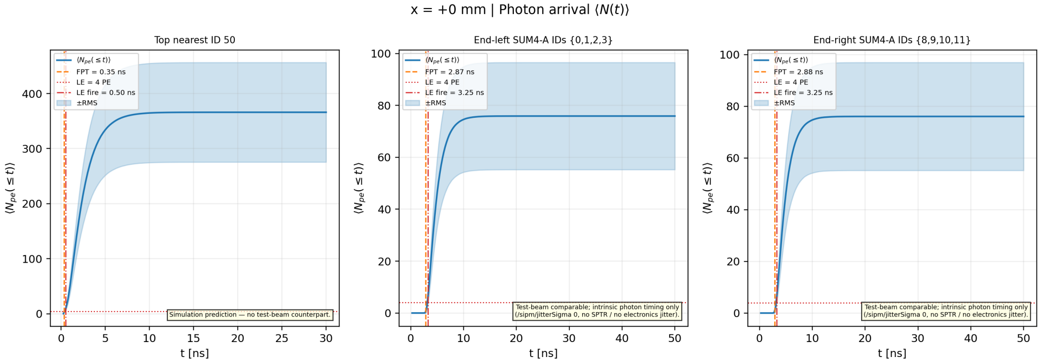
Position x=0 mm



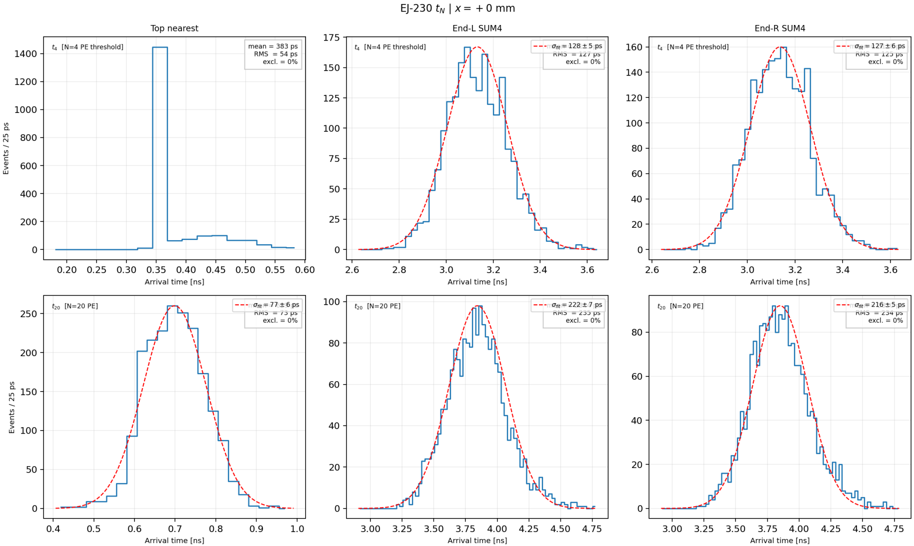


Quantity	Value
End L N_{pe}	151.0 ± 0.9
End R N_{pe}	151.1 ± 0.9
Nearest Top ID	50 (exact geometry)
Nearest Top N_{pe}	366.0 ± 2.0
Nearest Top $\langle t \rangle$	2.5528 ± 0.0020 ns
Nearest Top FPT	0.35099 ± 0.00023 ns

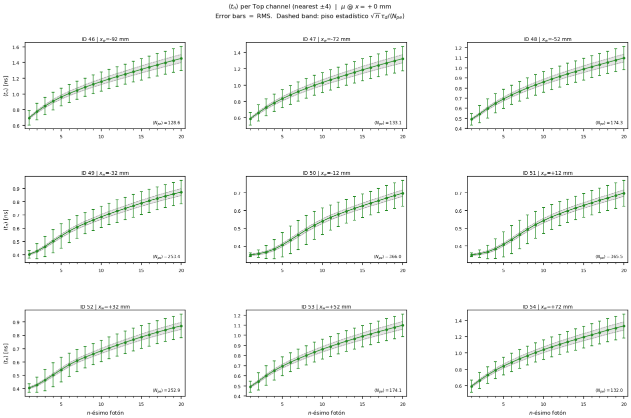
Run 4 | x = +0 mm | Photon arrival $\langle N(t) \rangle$

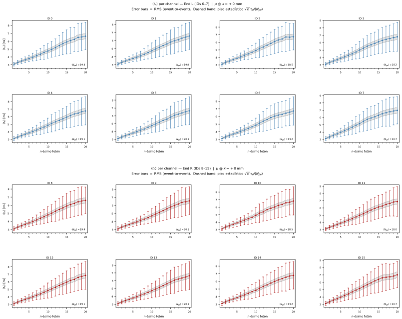


Run 4 | x = +0 mm | Time-to-threshold t_N



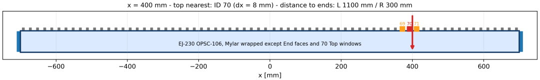
Run 4 | x = +0 mm | $\langle t_n \rangle$ per channel (dispersion)

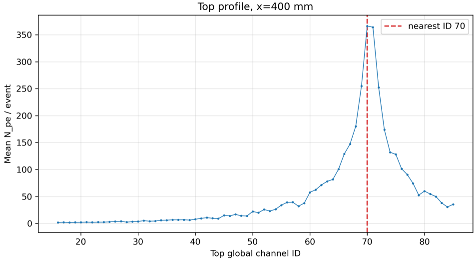




Error bars = RMS (event-to-event). Dashed band: statistical floor $\sigma_{stat}(t_n) \approx \sqrt{n}\,\tau_d/\langle N_{pe}\rangle$ ($\tau_d = 1.5$ ns). See dispersion-decomposition frame for physical interpretation.

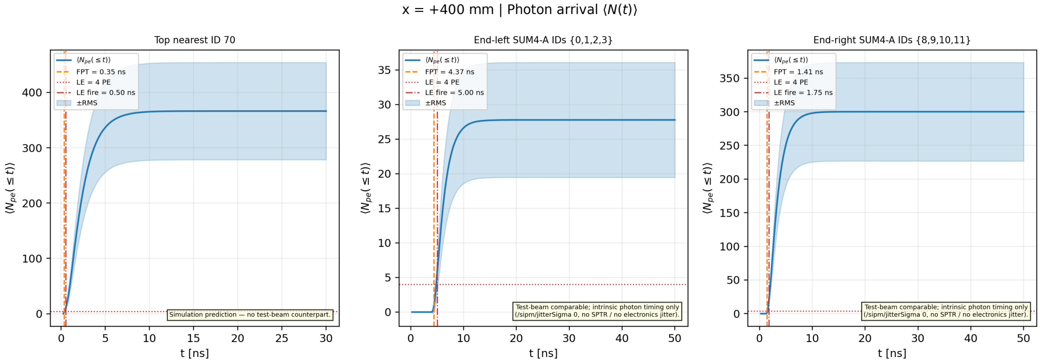
Position x=400 mm



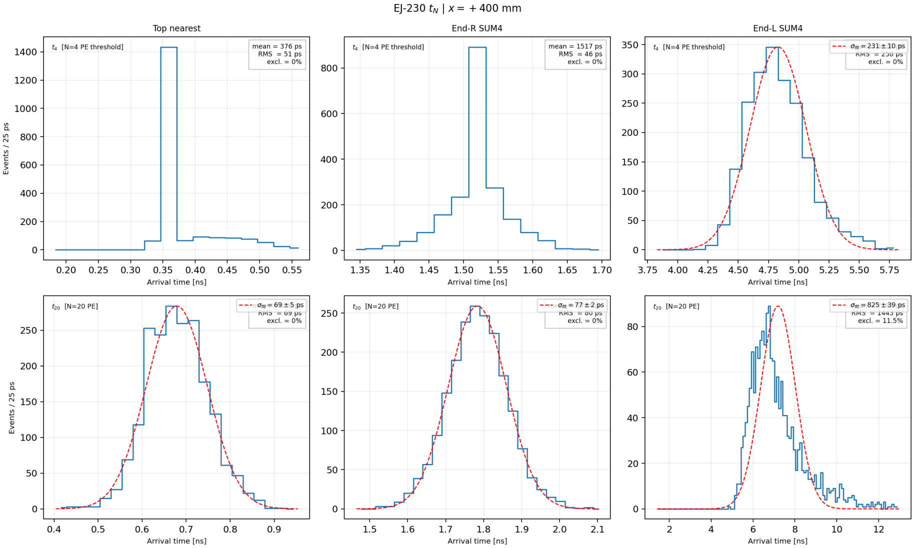


Quantity	Value
End L N_{pe}	52.8 ± 0.3
End R N_{pe}	577.8 ± 3.1
Nearest Top ID	70 (exact geometry)
Nearest Top N_{pe}	366.3 ± 2.0
Nearest Top $\langle t \rangle$	2.4909 ± 0.0020 ns
Nearest Top FPT	0.34615 ± 0.00025 ns

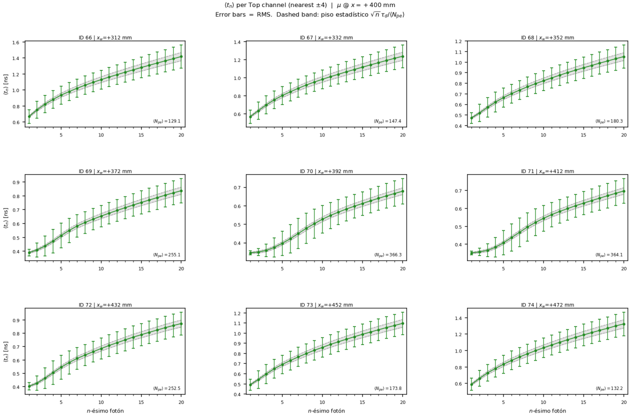
Run 5 | x = +400 mm | Photon arrival $\langle N(t) \rangle$

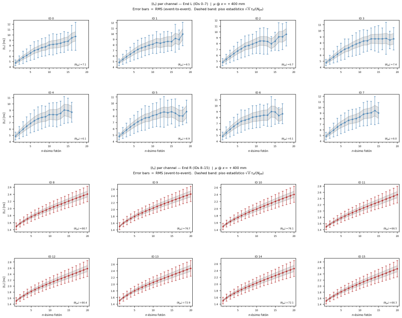


Run 5 | x = +400 mm | Time-to-threshold t_N



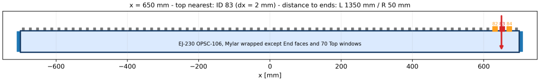
Run 5 | x = +400 mm | $\langle t_n \rangle$ per channel (dispersion)

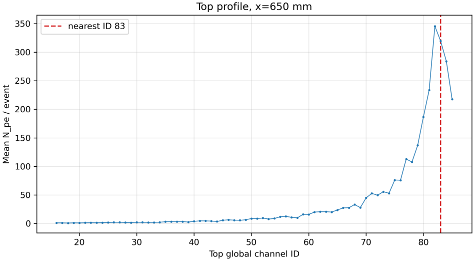




Error bars = RMS (event-to-event). Dashed band: statistical floor $\sigma_{stat}(t_n) \approx \sqrt{n}\,\tau_d/\langle N_{pe}\rangle$ ($\tau_d = 1.5$ ns). See dispersion-decomposition frame for physical interpretation.

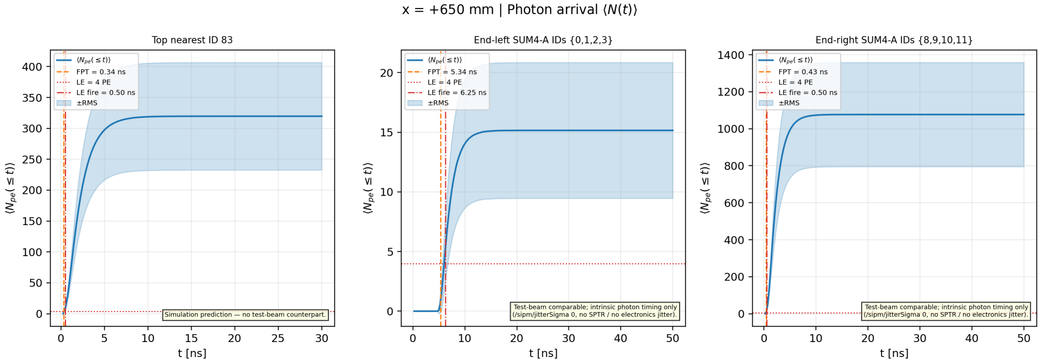
Position x=650 mm



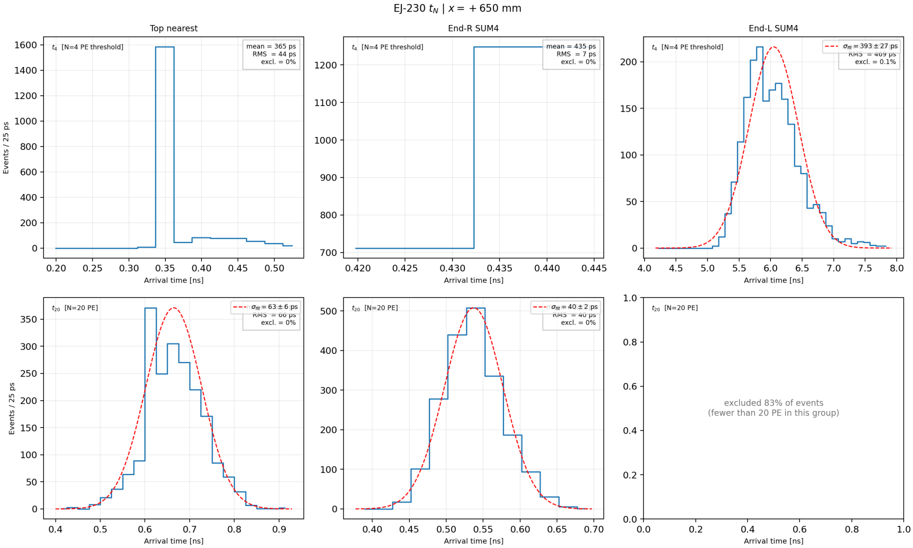


Quantity	Value
End L N_{pe}	30.2 ± 0.2
End R N_{pe}	2133.3 ± 12.4
Nearest Top ID	83 (window-track exception)
Nearest Top N_{pe}	319.7 ± 1.9
Nearest Top $\langle t \rangle$	2.3425 ± 0.0020 ns
Nearest Top FPT	0.34249 ± 0.00015 ns

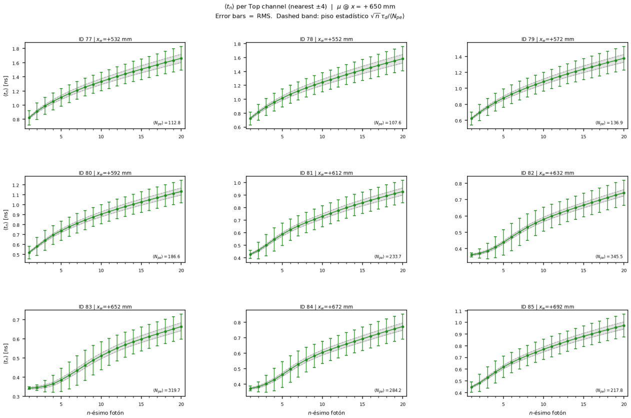
Run 6 | x = +650 mm | Photon arrival $\langle N(t)\rangle$

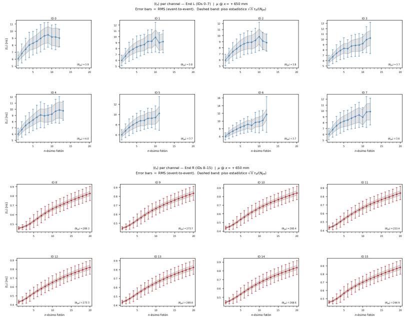


Run 6 | x = +650 mm | Time-to-threshold t_N



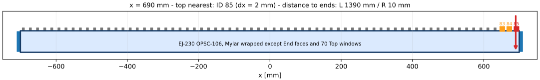
Run 6 | x = +650 mm | $\langle t_n \rangle$ per channel (dispersion)

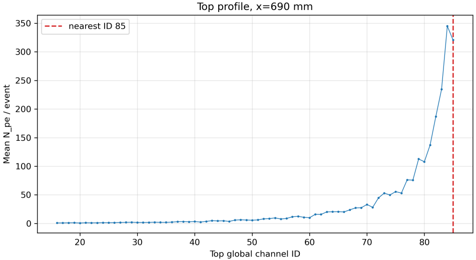




Error bars = RMS (event-to-event). Dashed band: statistical floor $\sigma_{stat}(t_n) \approx \sqrt{n}\,\tau_d/\langle N_{pe}\rangle$ ($\tau_d = 1.5$ ns). See dispersion-decomposition frame for physical interpretation.

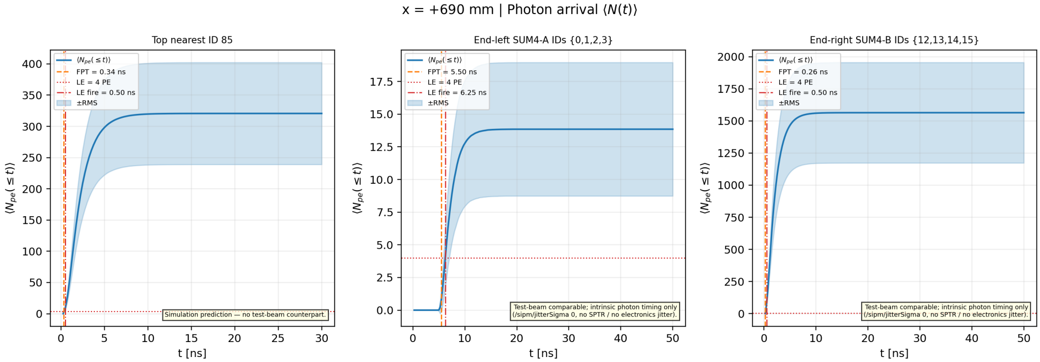
Position x=690 mm



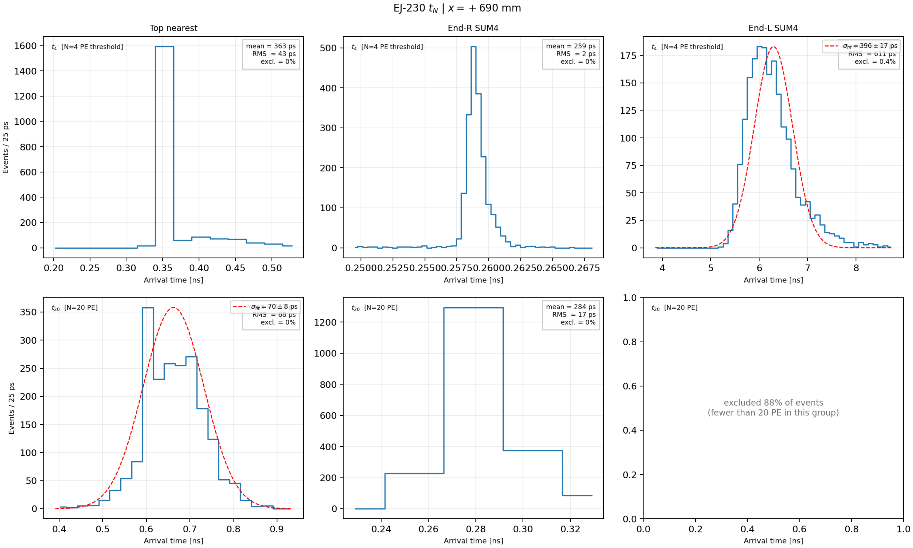


Quantity	Value
End L N_{pe}	27.6 ± 0.2
End R N_{pe}	3089.6 ± 17.1
Nearest Top ID	85 (window-track exception)
Nearest Top N_{pe}	320.6 ± 1.8
Nearest Top $\langle t \rangle$	2.3455 ± 0.0020 ns
Nearest Top FPT	0.34224 ± 0.00015 ns

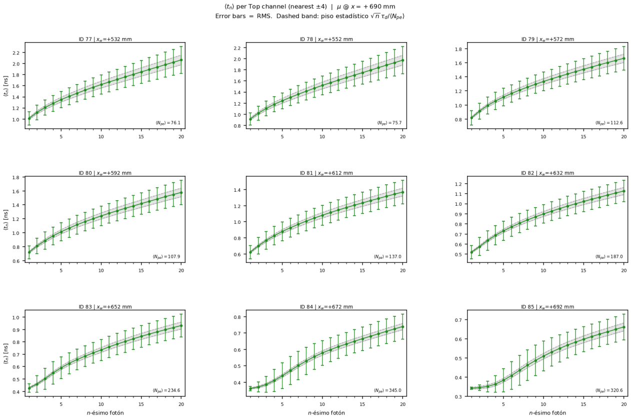
Run 7 | x = +690 mm | Photon arrival $\langle N(t) \rangle$

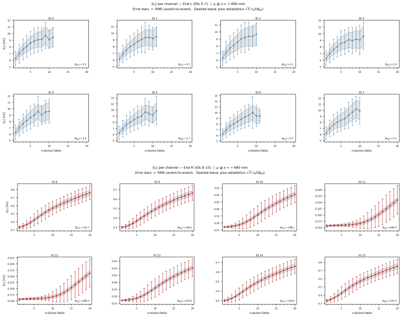


Run 7 | x = +690 mm | Time-to-threshold t_N



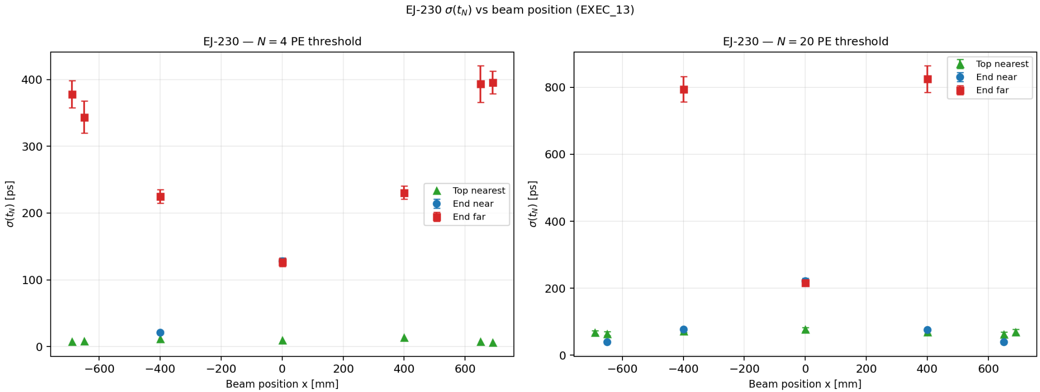
Run 7 | x = +690 mm | $\langle t_n \rangle$ per channel (dispersion)





Error bars = RMS (event-to-event). Dashed band: statistical floor $\sigma_{stat}(t_n) \approx \sqrt{n}\,\tau_d/\langle N_{pe}\rangle$ ($\tau_d = 1.5$ ns). See dispersion-decomposition frame for physical interpretation.

$\sigma(t_N)$ vs beam position: synthesis



CSV-derived synthesis

Near-End RMS(t_4) spans 2–127 ps across key positions; nearest-Top RMS(t_4) spans 43–54 ps. At $x = 0$, the two End groups give 127 ps.

Why does t_n disperse? Decomposition into four mechanisms

1. **Order statistics on the emission tail (irreducible):** t_n are order statistics of N_{pe} draws from the emission pdf ($\tau_r = 0.5$ ns, $\tau_d = 1.5$ ns); successive gaps $\sim \mathrm{Exp}(\tau_d/(N_{pe} - n + 1))$, $\sigma_{\mathrm{stat}}(t_n) \approx \sqrt{n}\,\tau_d/N_{pe}$ for $n \ll N_{pe}$ — grows monotonically with n even with perfect optics.
2. **Landau N_{pe} fluctuations:** $\mathrm{Var_L}(t_n) = (\partial\langle t_n\rangle/\partial N_{pe})^2\,\mathrm{Var}(N_{pe})$ with $\mathrm{Var}(N_{pe})$ from the documented $F = 1 + c\langle N_{pe}\rangle$. Statistical but non-Poissonian.
3. **Optical path-length dispersion:** TIR photons at $n = 1.58$ ($\theta_c \approx 39^\circ$) span axial speeds $c/(n\cos\theta) \to 1.4$–1.5 ns/m per metre plus Mylar bounce tail ($R = 0.98$). Low-n photons: quasi-direct; high-n: reflection-dominated $\Rightarrow$ RMS $\propto n$.
4. **Attenuation truncation:** distant channels ($\lambda_{\mathrm{eff}} \approx 30.8$ cm) populate only $n = 1$–3 with large purely statistical bars.

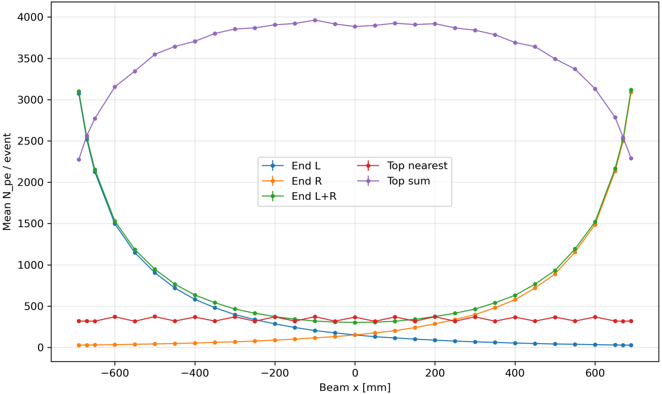
Takeaway

$\mathrm{RMS}(t_n) - \sqrt{n}\,\tau_d/N_{pe}$ isolates the optical-propagation contribution — the per-channel diagnostic in the $\langle t_n\rangle$ dispersion frames and the same dispersion that limits σ_{group}.

Simulation only / intrinsic

Top: simulation prediction, no test-beam counterpart. End: test-beam comparable, intrinsic precision (no SPTR / no jitter).

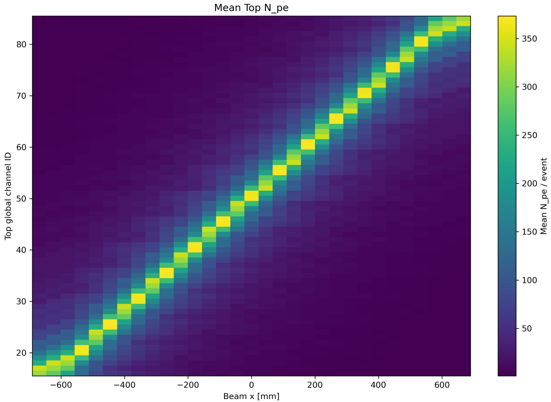
Photon budget versus beam position



How to read this plot

Why: Quantify N_{pe} collected at each End and nearest-Top channel as the muon track scans the bar.
Axes: x: beam position [mm]; y: mean N_{pe} per event (End sums left/right and nearest Top).
Takeaway: N_{pe} decreases exponentially with track-to-End distance ($\lambda_{eff} \approx 30.8$ cm $\ll$ ABSLENGTH $= 120$ cm, system effective length); nearest-Top N_{pe} peaks directly under the SiPM window.

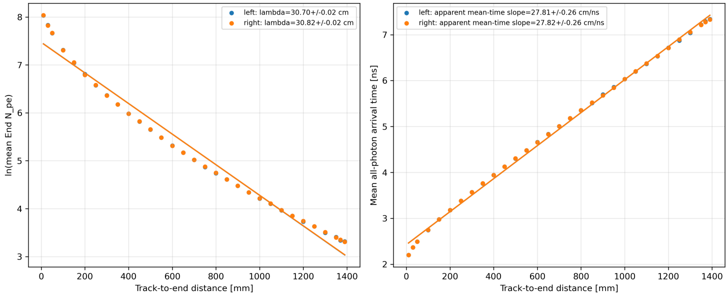
Top localization: full coverage diagonal



CSV-derived localization gate

31/31 positions pass. 18 positions use the documented window-track rule; the maximum nearest-channel deficit is 8.37%. All remaining positions have their strict maximum at the geometrically nearest channel.

Attenuation and the historical mean-time slope



$\lambda_{\mathrm{eff},L} = 30.70 \pm 0.02$ cm, $\lambda_{\mathrm{eff},R} = 30.82 \pm 0.02$ cm; the mean-time slope is tail-biased and is not a propagation velocity.

How to read this plot

Why: Measure the effective attenuation length λ_{eff} and test whether the mean-time slope encodes a light-propagation velocity.

Axes: Left: mean N_{pe} vs track-to-End distance [cm]; exponential fit gives λ_{eff}. Right: mean arrival time [ns] vs same distance.

Takeaway: $\lambda_{eff} \approx 30.8$ cm $\ll$ ABSLENGTH $= 120$ cm: λ_{eff} is a system effective length combining bulk absorption, Mylar reflection losses, and Top window drainage—not the material property. The mean-time slope is tail-biased; no propagation velocity is reliably extracted.

Why the Fano factor grows with $\langle N_{pe} \rangle$

1. **Pure Poisson (fixed ΔE):** $\mathrm{Var}(N_{pe}) = \langle N_{pe} \rangle$, hence $F \equiv 1$.
2. **Landau fluctuations in ΔE** (law of total variance):

$$\mathrm{Var}(N_{pe}) = \langle N_{pe} \rangle + \left(\frac{\langle N_{pe} \rangle}{\langle \Delta E \rangle} \right)^2 \mathrm{Var}(\Delta E),$$

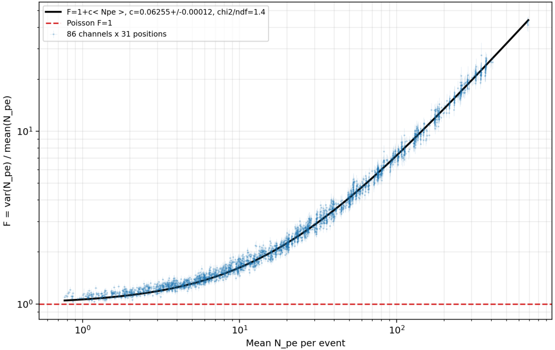
   $\Rightarrow \; F = 1 + c \langle N_{pe} \rangle, \; c = \mathrm{Var}(\Delta E) / \langle \Delta E \rangle^2$ (collection-efficiency terms absorbed in c).
3. **Poisson limit $F \to 1$:** only at $\langle N_{pe} \rangle \to 0$ (counting noise dominates over energy-loss fluctuations).
4. **Effective relative fluctuation:** $\sqrt{c} \approx 0.250$ is the combined energy-loss + collection jitter, consistent with a thin (10 mm) plastic absorber.

Fit result (see next slide)

$c = 0.062547 \pm 0.000117$

The *same* Landau ΔE fluctuation that shapes the asymmetric N_{pe} spectrum (next section) also generates the super-Poissonian Fano factor: **one physical mechanism, two observables**.

Landau dominance: Fano factor versus mean N_{pe}



$F = 1 + c\langle N_{pe}\rangle$, $c = 0.062547 \pm 0.000117$, $\chi^2/\mathrm{ndf} = 1.41$. Poisson $F = 1$ is approached only at low light.

How to read this plot

Why: Measure $F = \mathrm{Var}(N_{pe})/\langle N_{pe}\rangle$ to distinguish Poisson ($F = 1$) from Landau-dominated fluctuations.
Axes: x: $\langle N_{pe}\rangle$ per event; y: $\mathrm{Var}/\langle N_{pe}\rangle$. Dashed: Poisson $F = 1$. Line: linear fit $F = 1 + c\langle N_{pe}\rangle$.
Takeaway: F grows linearly: Landau ΔE fluctuations dominate counting noise at all light levels. $\sqrt{c}$ is the effective relative energy-loss fluctuation of this 10 mm absorber.

Why Landau (and why Moyal as analytic proxy)

Why not Gaussian:

- In 10 mm of plastic the muon undergoes many collisions, but the energy-transfer cross-section ($\mathrm{d}\sigma/\mathrm{d}\varepsilon \propto 1/\varepsilon^2$, Rutherford/Møller) has a hard tail.
- Rare large transfers (δ-rays) dominate the variance; the variance per collision diverges and the Central Limit Theorem does not apply.
- Result: asymmetric distribution with a hard right tail — the Landau distribution.
- Gaussian (Vavilov→Gauss) limit only applies to *thick* absorbers.

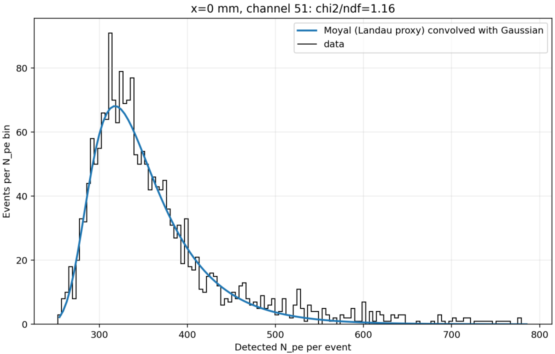
Why Moyal:

- Landau density has no closed form (contour integral).
- The Moyal function is a closed-form analytic approximation to the Landau core and early tail, enabling stable MINUIT fits.
- *Known limitation:* extreme Moyal tail undershoots true Landau tail; fit quality degrades at high N_{pe}.

Moyal $\otimes$ Gauss:

- Photo-electron counting (Poisson) and collection fluctuations broaden the peak symmetrically.
- Full model: Moyal $\otimes$ Gaussian.
- Connects to Fano slide: same Landau ΔE fluctuation, two observables (spectrum shape + super-Poissonian F).

High-light example: Moyal–Gaussian Landau proxy

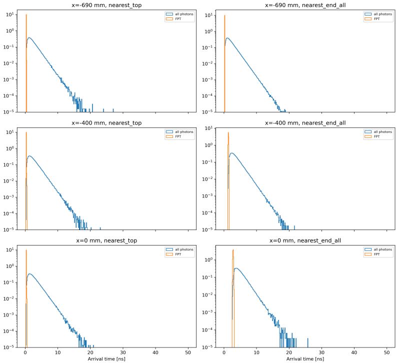


Moyal approximates Landau. Some channels have poor χ^2 or an unidentifiable Gaussian width; all recorded in `exec10_landau_mpv.csv`.

How to read this plot

Why: Fit the N_{pe} spectrum at a high-yield position to quantify the Landau MPV and width.
Axes: x: N_{pe} per event; y: counts. Data (step), Moyal$\otimes$Gauss fit (curve).
Takeaway: Core described by Moyal proxy; hard right tail from δ-rays. MPVs are future inputs for test-beam ToT calibration.

Arrival-time examples: absolute rate and normalized shape

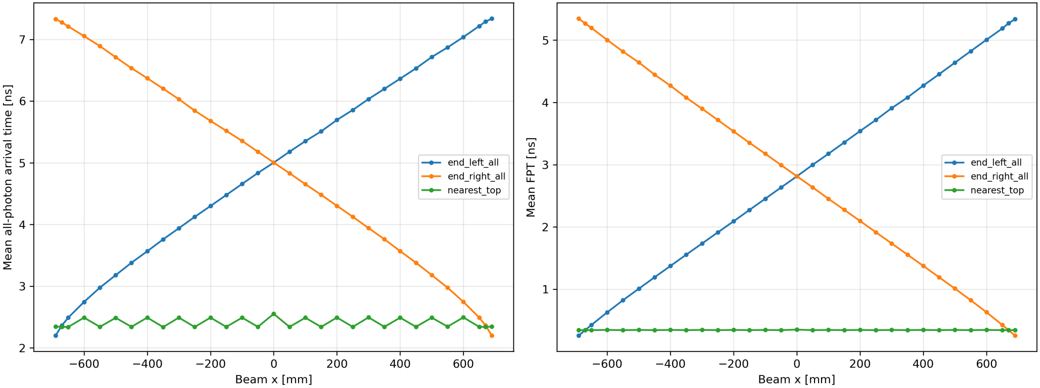


Left: entries per event per ns. Right: area-normalized distributions. FPT = first detected photon time per event.

How to read this plot

Why: Illustrate the arrival-time structure of individual Top channels and contrast absolute rate vs shape.
Axes: x: arrival time [ns]; y: left is entries/event/ns, right is area-normalized density [1/ns].
Takeaway: FPT distribution captures the prompt scintillation leading edge; the broad tail reflects multiple-reflection photons.

Mean arrival time and FPT versus x



How to read this plot

Why: Track how the mean photon arrival time and FPT scale with beam position.
Axes: x: beam position [mm]; y: mean arrival time [ns] and mean FPT [ns].
Takeaway: Both estimators increase with track-to-End distance; the mean is more tail-biased than FPT and yields a larger apparent slope.

Nearest-Top Landau/Moyal MPVs at key positions

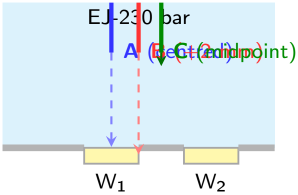
x [mm]	channel	MPV [Npe]	Landau width [Npe]	χ^2/ndf
-690	16	275.0 ± 0.9	23.3 ± 0.8	1.61
-650	18	275.6 ± 0.9	23.7 ± 0.8	1.58
-400	31	317.5 ± 1.0	26.8 ± 0.9	1.56
+0	50	318.5 ± 1.0	26.0 ± 0.9	1.17
+400	70	320.1 ± 1.0	25.3 ± 0.8	1.23
+650	83	275.0 ± 0.9	24.7 ± 0.8	1.87
+690	85	278.1 ± 0.9	24.6 ± 0.8	2.15

Interpretation

The right tail originates primarily from Landau-like energy-loss fluctuations; Poisson counting dominates only in the few-PE limit.

Window-track collection effect: mechanism

- Top SiPMs couple through index-matched windows (6×6 mm^2) in the +Y Mylar face.
- First-pass photon collection depends on the solid angle subtended by the window from the muon track (geometric working hypothesis, consistent with five-run pattern).
- Empirically, a track at ± 2 mm from the window centre changes the local contrast by 7.9% with minimum significance 9.3σ. The measured centered and midpoint contrast significances are -1.02σ and -0.06σ, respectively.
- Scan positions with $x \equiv 10 \pmod{20}$ mm fall at ± 2 mm from a window centre — hence the "(window-track exception)" label in the tables.
- Effect is local, symmetric, characterised and bounded: localization diagonal and channel map remain valid.

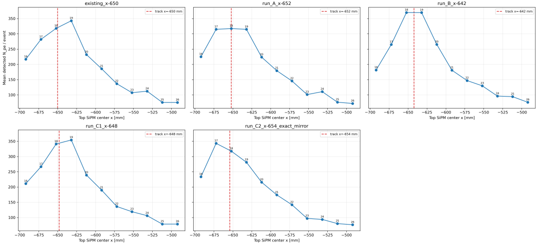


Consequence for the scan

Track B subtends a larger first-pass solid angle to W$_1$; fewer multiple reflections $\Rightarrow$ higher N_{pe} at the nearest channel. At midpoint (C) both windows contribute equally and the effect is null.

Window-track collection effect: five-run test

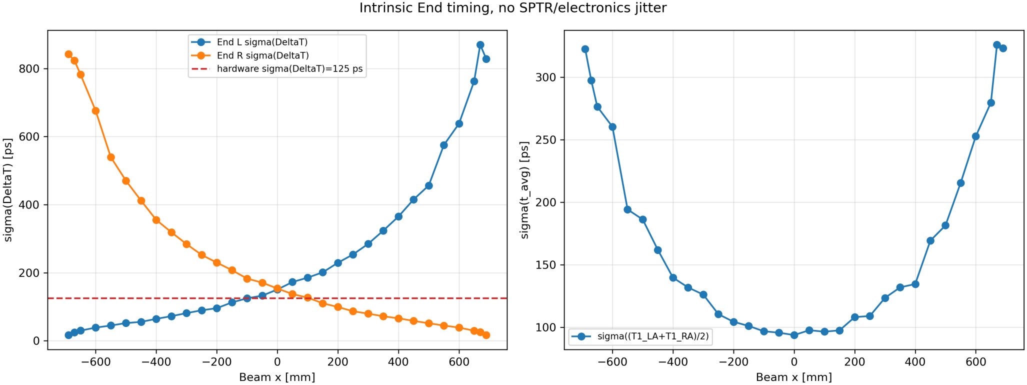
Run	contrast [Npe]	significance
Existing -650	24.96 ± 2.57	9.72
A -652 center	-2.44 ± 2.39	-1.02
B -642 midpoint	-0.19 ± 3.04	-0.06
C1 -648 (+4 mm)	13.27 ± 2.73	4.85
C2 -654 mirror	25.54 ± 2.76	9.26



Finding

Local symmetric ± 2 mm effect: 7.9% with significance 9.3–9.7σ; centered/midpoint contrast significances: -1.02σ/-0.06σ.

Intrinsic End SUM4 timing



$\sigma_{group} = \sigma(\Delta T_{AB})/\sqrt{2}$; no SPTR or electronics jitter.

CSV-derived timing summary

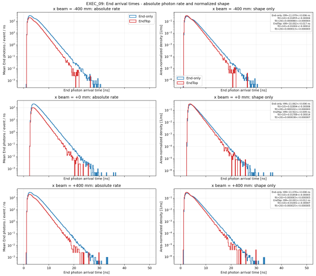
Intrinsic End σ_{group} at $x = 0$: 107.6 ps; full-scan mean 177 ps and range 12–615 ps.

End arrival-time tails without any time acceptance window: narrower EndTop is physical

x	side	EndTop [ps]	End-only [ps]	ratio
-400	L	45.5 ± 0.7	37.2 ± 0.3	1.223 ± 0.021
-400	R	251.3 ± 4.0	128.0 ± 0.9	1.963 ± 0.034
+0	L	106.5 ± 1.7	76.4 ± 0.5	1.394 ± 0.024
+0	R	108.7 ± 1.7	75.3 ± 0.5	1.443 ± 0.025
+400	L	258.2 ± 4.1	131.5 ± 0.9	1.963 ± 0.034
+400	R	46.7 ± 0.7	37.9 ± 0.3	1.233 ± 0.021

Confirmed mechanism

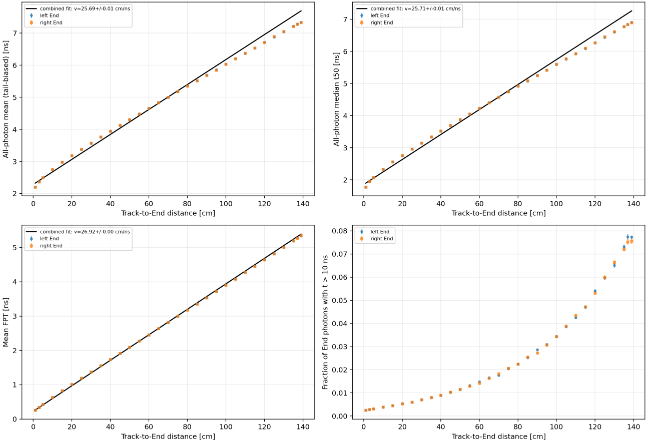
Across $x = 0, \pm 400$ mm, EndTop minus End-only t_{99} ranges from -1.08 to -0.92 ns; the minimum absolute significance among the two late-fraction differences is 6.6σ.



No time acceptance window

All End-photon arrivals are used without a $t < t_{max}$ cut. EndTop tail narrowing persists over the full range: physical, not an artefact.

Apparent timing slopes: no propagation estimator passes



Estimator	slope-derived [cm/ns]	χ^2/ndf
all-photon mean	25.694 ± 0.005	3357.5
all-photon t_{50}	25.714 ± 0.005	4143.2
mean FPT	26.922 ± 0.003	742.6

Refuted predictions

Every global line fit is rejected; slope-derived values are not accepted propagation velocities.

Top readout: analytic timing estimate only

x [mm]	N_{pe}	$\sigma_{t,est}$ [ps]
-690	318.7 ± 1.8	48.51 ± 0.14
-650	318.0 ± 1.8	48.57 ± 0.14
-400	367.2 ± 2.1	45.19 ± 0.13
+0	366.0 ± 2.0	45.27 ± 0.13
+400	366.3 ± 2.0	45.25 ± 0.12
+650	319.7 ± 1.9	48.44 ± 0.15
+690	320.6 ± 1.8	48.36 ± 0.14

Definition, derivation, and scope

$\sigma_{t,est} = \sqrt{\tau_r \tau_d}/\sqrt{N_{pe}}$. Threshold at low N_{pe}, double-exponential emission; analytic orientation only.

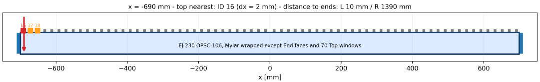
$\sigma_{t,est}$ vs. simulated $\sigma(t_4)$

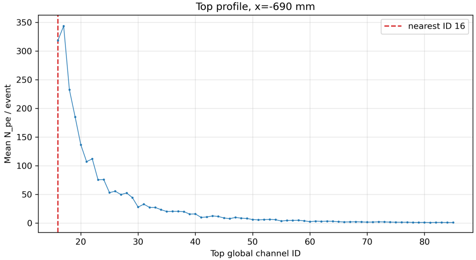
x	$\sigma_{t,est}$ [ps]	fitted $\sigma(t_4)$ [ps]
-690	48.51 ± 0.14	7.8
-650	48.57 ± 0.14	8.3
-400	45.19 ± 0.13	11.4
+0	45.27 ± 0.13	9.5
+400	45.25 ± 0.12	13.5
+650	48.44 ± 0.15	7.8
+690	48.36 ± 0.14	6.0

Numerical conclusions

- At center: End sum 302.1, nearest Top 366.0; at $x = -690$ mm: End_L 3073.3, nearest Top 318.7.
- λ_{eff}: L 30.70±0.02 cm, R 30.82±0.02 cm.
- Nearest-Top $F = 1 + (0.062547 \pm 0.000117)\langle N_{pe}\rangle$ ($\chi^2/ndf = 1.41$).
- Intrinsic End σ_{group} at $x = 0$: 107.6 ps; full-scan mean 177 ps and range 12–615 ps.
- EndTop/End-only timing ratio: 1.223–1.963; no time-acceptance window.
- Near-End RMS(t_4): 2–127 ps; nearest-Top RMS(t_4): 43–54 ps.
- Directed EJ-230 window and End-only controls replace all inherited special-control placeholders.

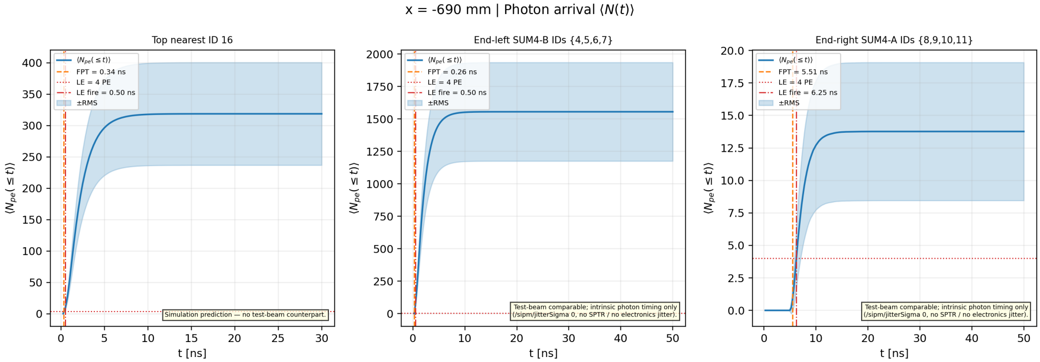
Position x=-690 mm



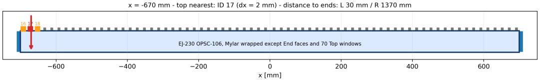


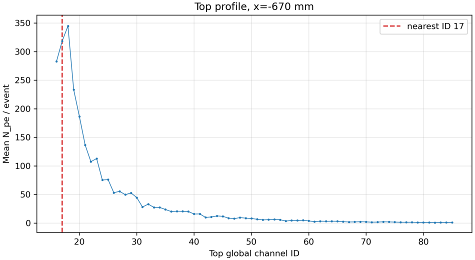
Quantity	Value
End L N_{pe}	3073.3 ± 16.8
End R N_{pe}	27.4 ± 0.2
Nearest Top ID	16 (window-track exception)
Nearest Top N_{pe}	318.7 ± 1.8
Nearest Top $\langle t \rangle$	2.3460 ± 0.0020 ns
Nearest Top FPT	0.34237 ± 0.00012 ns

Run 1 | x = -690 mm | Photon arrival $\langle N(t) \rangle$



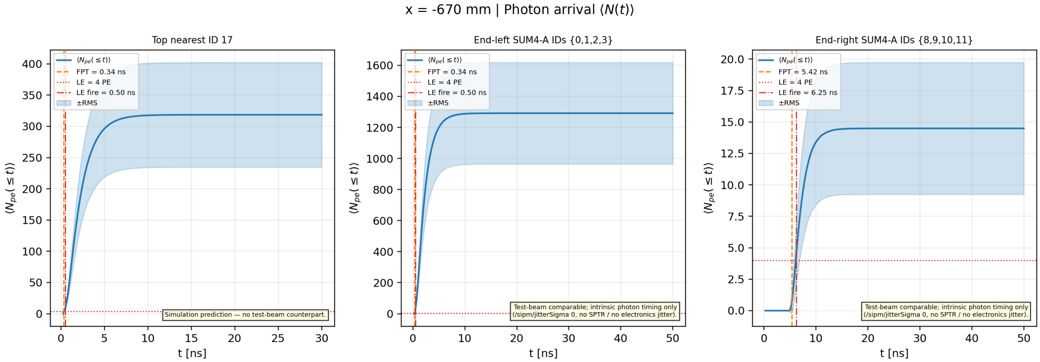
Position x=-670 mm



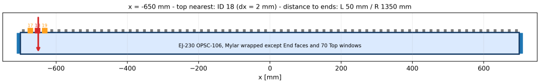


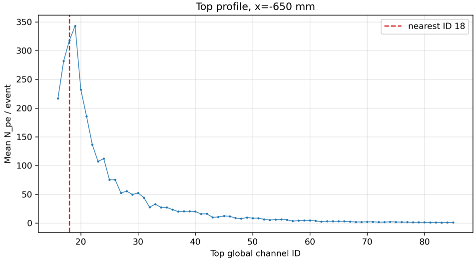
Quantity	Value
End L N_{pe}	2516.7 ± 14.2
End R N_{pe}	28.6 ± 0.2
Nearest Top ID	17 (window-track exception)
Nearest Top N_{pe}	318.5 ± 1.9
Nearest Top $\langle t \rangle$	2.3425 ± 0.0020 ns
Nearest Top FPT	0.34229 ± 0.00020 ns

Run 2 | x = -670 mm | Photon arrival $\langle N(t) \rangle$



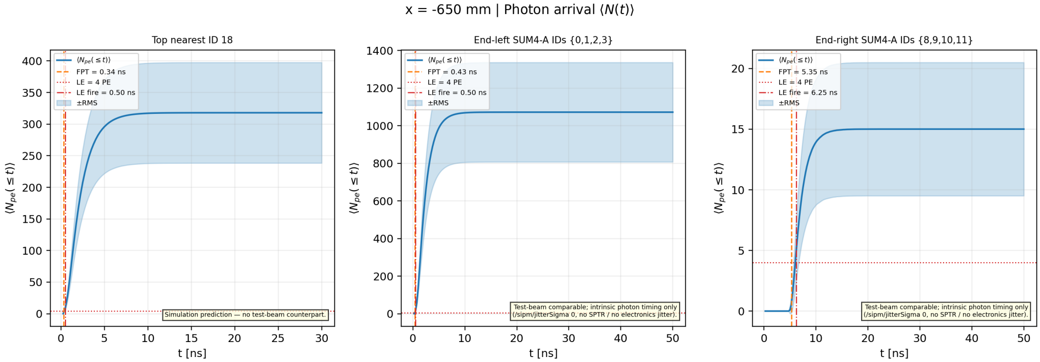
Position x=-650 mm



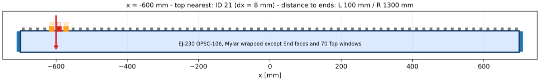


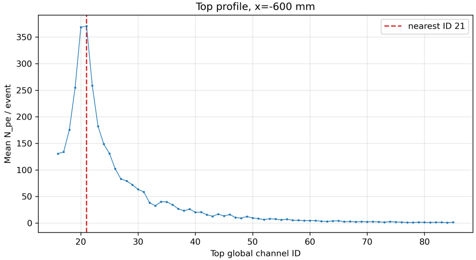
Quantity	Value
End L N_{pe}	2124.3 ± 11.6
End R N_{pe}	29.9 ± 0.2
Nearest Top ID	18 (window-track exception)
Nearest Top N_{pe}	318.0 ± 1.8
Nearest Top $\langle t \rangle$	2.3394 ± 0.0020 ns
Nearest Top FPT	0.34248 ± 0.00020 ns

Run 3 | x = -650 mm | Photon arrival $\langle N(t) \rangle$



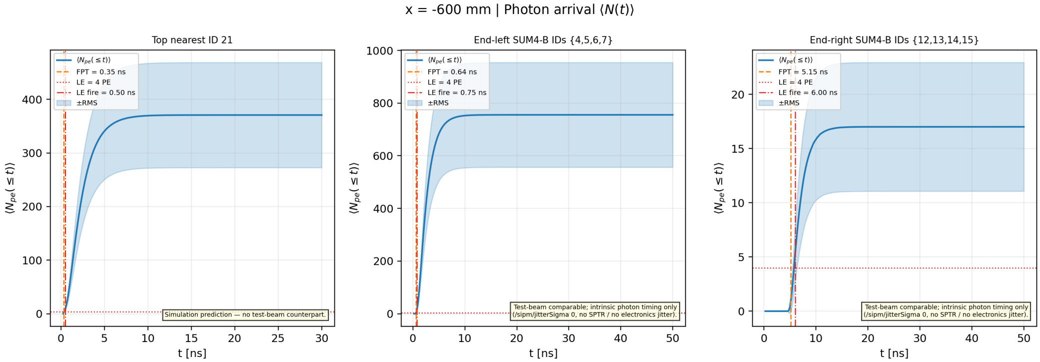
Position x=-600 mm



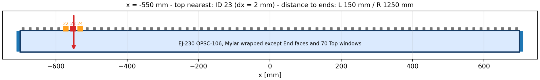


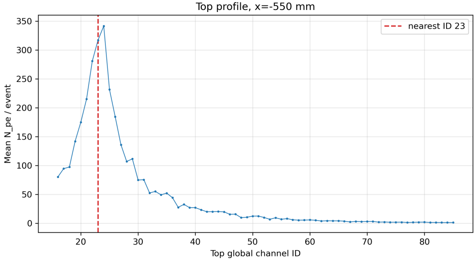
Quantity	Value
End L N_{pe}	1496.7 ± 8.8
End R N_{pe}	33.5 ± 0.2
Nearest Top ID	21 (exact geometry)
Nearest Top N_{pe}	370.7 ± 2.2
Nearest Top $\langle t \rangle$	2.4920 ± 0.0019 ns
Nearest Top FPT	0.34586 ± 0.00023 ns

Run 4 | x = -600 mm | Photon arrival $\langle N(t) \rangle$



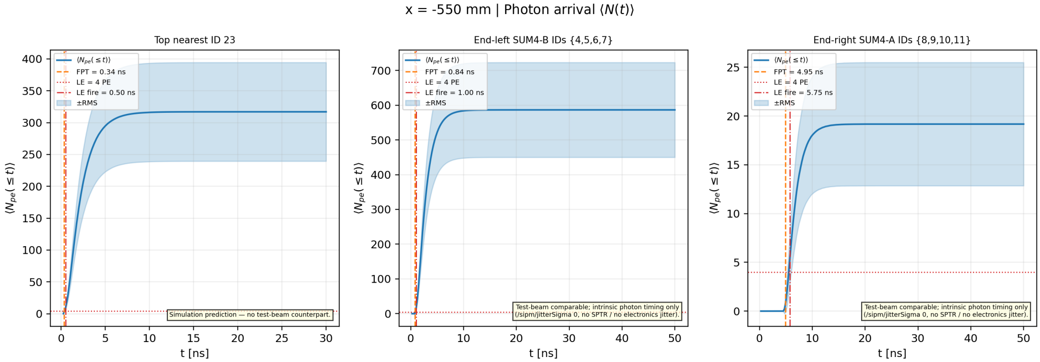
Position x=-550 mm



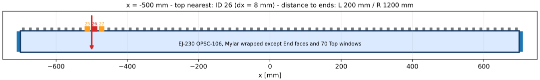


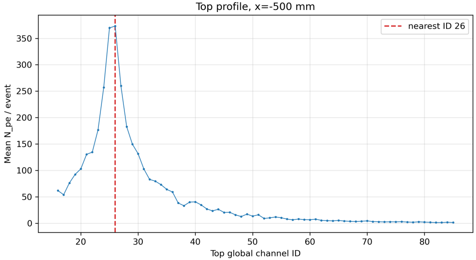
Quantity	Value
End L N_{pe}	1146.8 ± 6.0
End R N_{pe}	37.8 ± 0.2
Nearest Top ID	23 (window-track exception)
Nearest Top N_{pe}	316.9 ± 1.7
Nearest Top $\langle t \rangle$	2.3422 ± 0.0020 ns
Nearest Top FPT	0.34244 ± 0.00014 ns

Run 5 | x = -550 mm | Photon arrival $\langle N(t) \rangle$



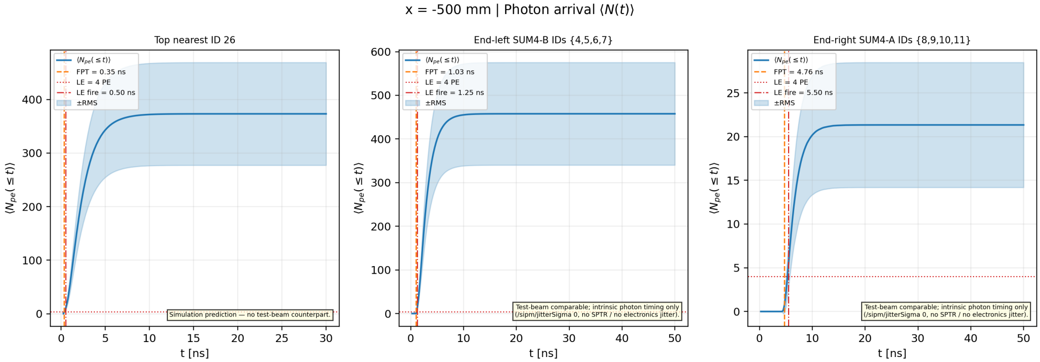
Position x=-500 mm



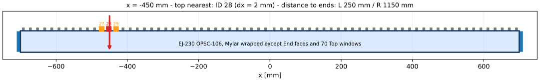


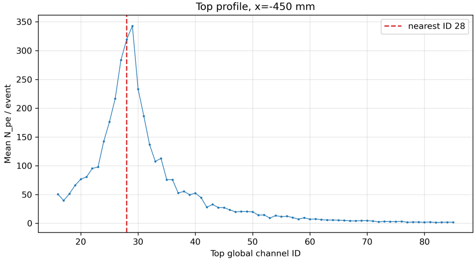
Quantity	Value
End L N_{pe}	904.4 ± 5.2
End R N_{pe}	42.3 ± 0.3
Nearest Top ID	26 (exact geometry)
Nearest Top N_{pe}	373.2 ± 2.1
Nearest Top $\langle t \rangle$	2.4886 ± 0.0019 ns
Nearest Top FPT	0.34538 ± 0.00014 ns

Run 6 | x = -500 mm | Photon arrival $\langle N(t) \rangle$



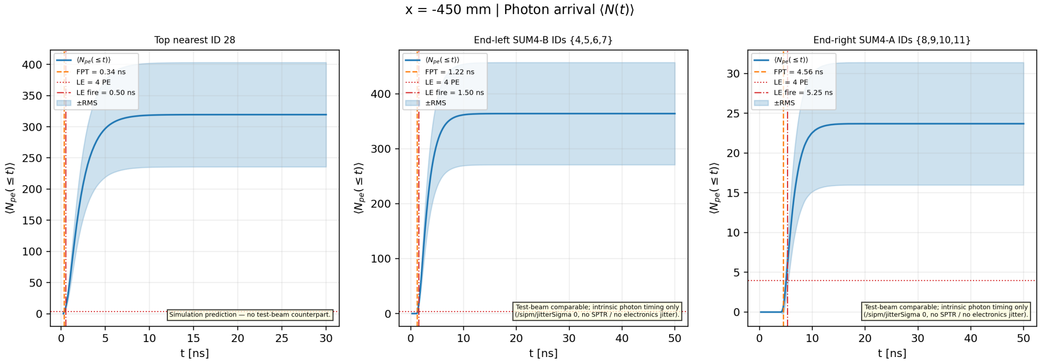
Position x=-450 mm



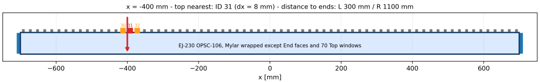


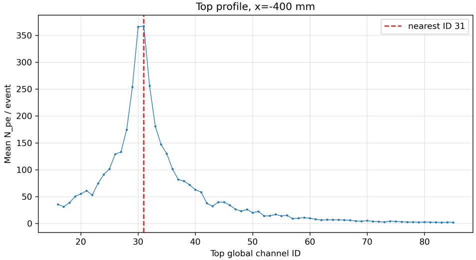
Quantity	Value
End L N_{pe}	718.6 ± 4.1
End R N_{pe}	47.0 ± 0.3
Nearest Top ID	28 (window-track exception)
Nearest Top N_{pe}	319.4 ± 1.9
Nearest Top $\langle t \rangle$	2.3423 ± 0.0020 ns
Nearest Top FPT	0.34239 ± 0.00020 ns

Run 7 | x = -450 mm | Photon arrival $\langle N(t) \rangle$



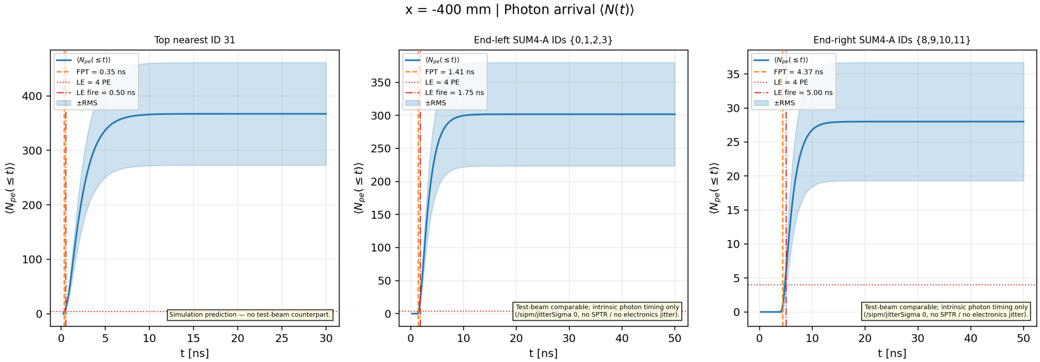
Position x=-400 mm



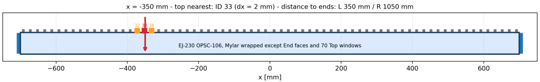


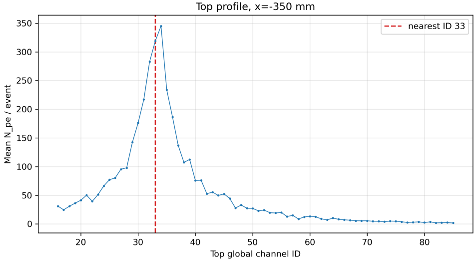
Quantity	Value
End L N_{pe}	581.1 ± 3.4
End R N_{pe}	53.0 ± 0.3
Nearest Top ID	31 (exact geometry)
Nearest Top N_{pe}	367.2 ± 2.1
Nearest Top $\langle t \rangle$	2.4926 ± 0.0020 ns
Nearest Top FPT	0.34578 ± 0.00019 ns

Run 8 | x = -400 mm | Photon arrival $\langle N(t) \rangle$



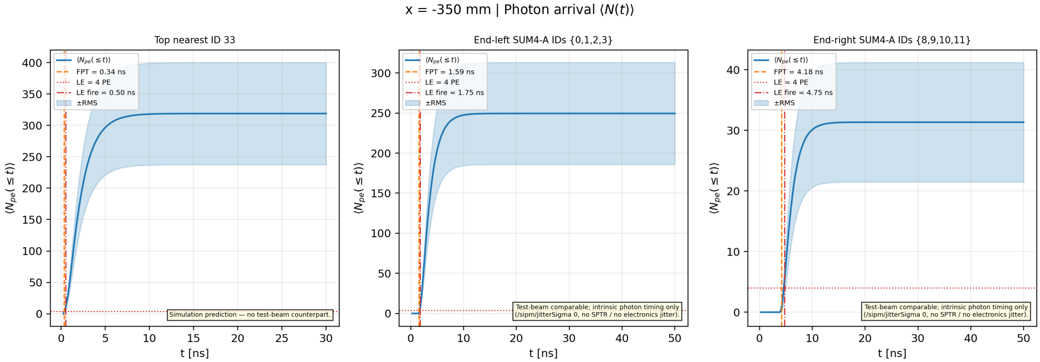
Position x=-350 mm



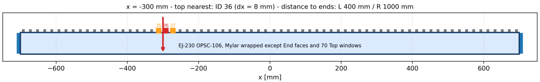


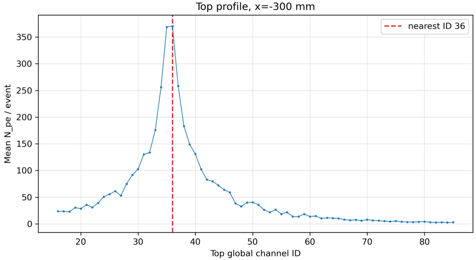
Quantity	Value
End L N_{pe}	481.1 ± 2.7
End R N_{pe}	60.7 ± 0.4
Nearest Top ID	33 (window-track exception)
Nearest Top N_{pe}	318.9 ± 1.8
Nearest Top $\langle t \rangle$	2.3390 ± 0.0020 ns
Nearest Top FPT	0.34245 ± 0.00015 ns

Run 9 | x = -350 mm | Photon arrival $\langle N(t) \rangle$



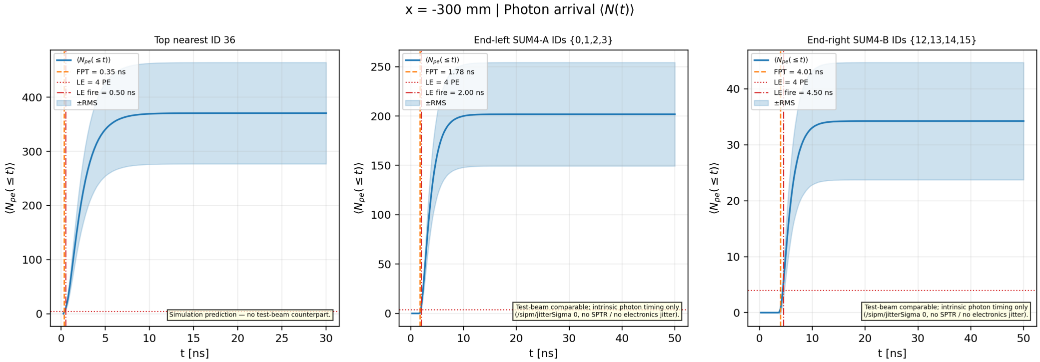
Position x=-300 mm



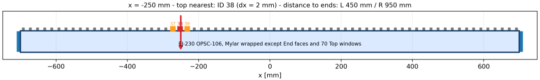


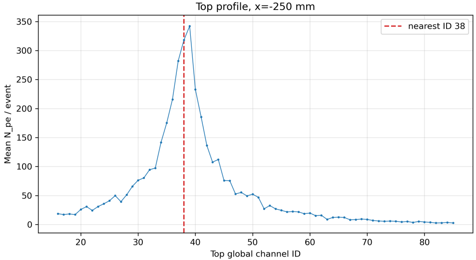
Quantity	Value
End L N_{pe}	397.1 ± 2.3
End R N_{pe}	67.6 ± 0.4
Nearest Top ID	36 (exact geometry)
Nearest Top N_{pe}	370.6 ± 2.1
Nearest Top $\langle t \rangle$	2.4914 ± 0.0019 ns
Nearest Top FPT	0.34579 ± 0.00017 ns

Run 10 | x = -300 mm | Photon arrival $\langle N(t) \rangle$



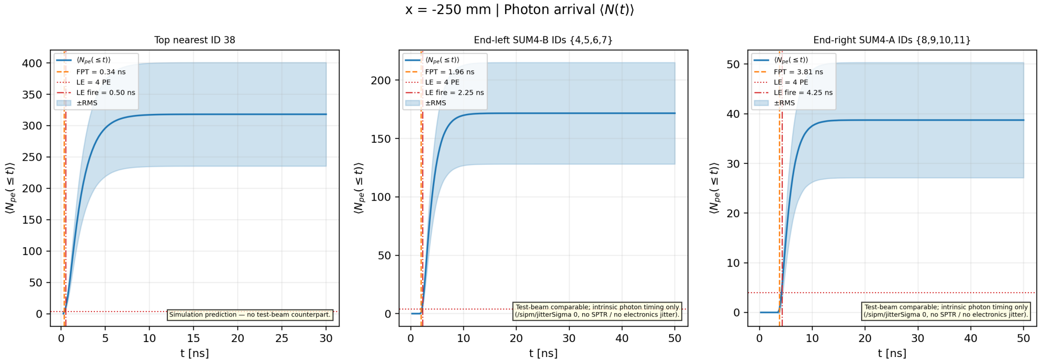
Position x=-250 mm



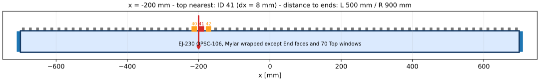


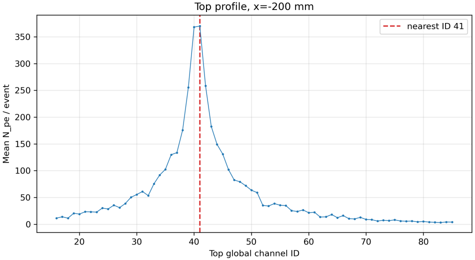
Quantity	Value
End L N_{pe}	336.8 ± 1.9
End R N_{pe}	76.8 ± 0.5
Nearest Top ID	38 (window-track exception)
Nearest Top N_{pe}	318.1 ± 1.8
Nearest Top $\langle t \rangle$	2.3401 ± 0.0020 ns
Nearest Top FPT	0.34234 ± 0.00017 ns

Run 11 | x = -250 mm | Photon arrival $\langle N(t) \rangle$



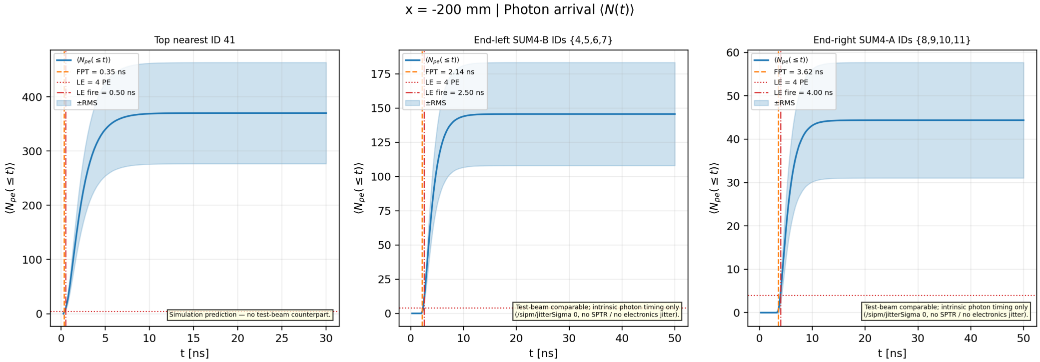
Position x=-200 mm



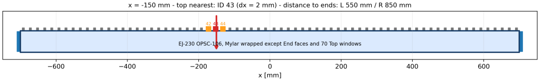


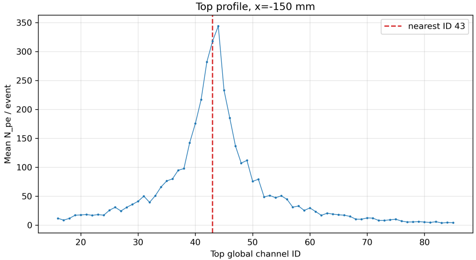
Quantity	Value
End L N_{pe}	285.2 ± 1.6
End R N_{pe}	87.9 ± 0.5
Nearest Top ID	41 (exact geometry)
Nearest Top N_{pe}	370.1 ± 2.1
Nearest Top $\langle t \rangle$	2.4913 ± 0.0019 ns
Nearest Top FPT	0.34568 ± 0.00014 ns

Run 12 | x = -200 mm | Photon arrival $\langle N(t) \rangle$



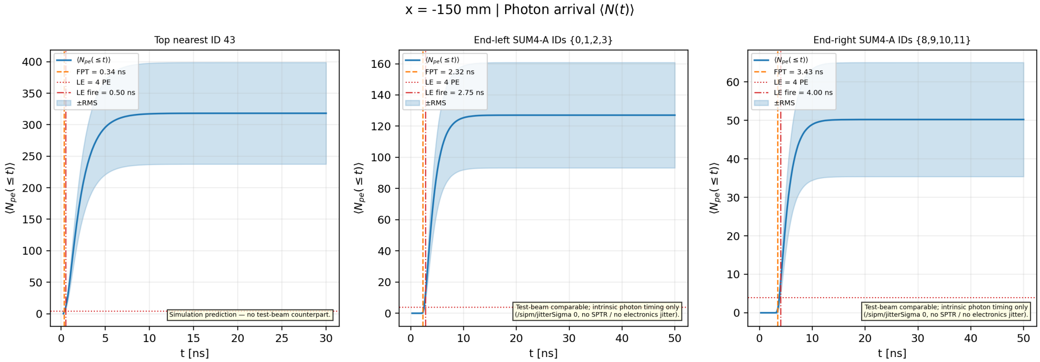
Position x=-150 mm



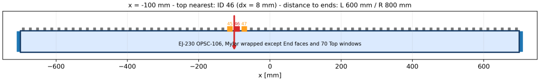


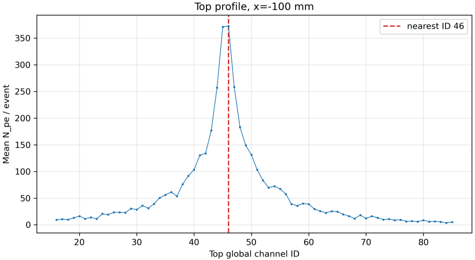
Quantity	Value
End L N_{pe}	241.0 ± 1.4
End R N_{pe}	100.4 ± 0.6
Nearest Top ID	43 (window-track exception)
Nearest Top N_{pe}	318.2 ± 1.8
Nearest Top $\langle t \rangle$	2.3419 ± 0.0020 ns
Nearest Top FPT	0.34245 ± 0.00019 ns

Run 13 | x = -150 mm | Photon arrival $\langle N(t) \rangle$



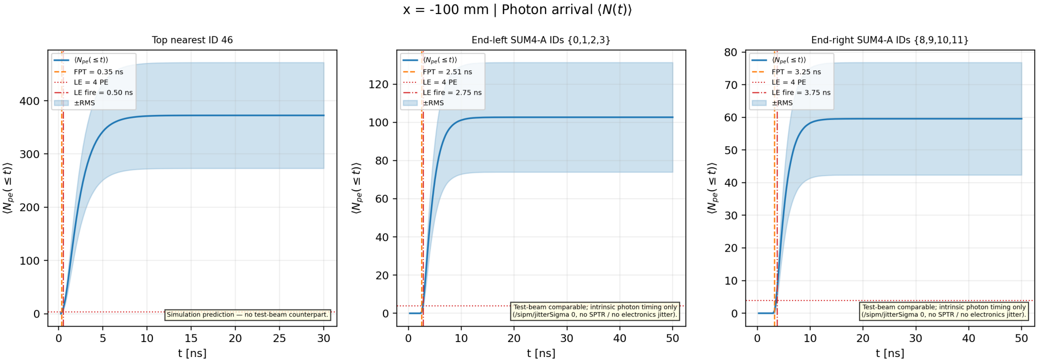
Position x=-100 mm



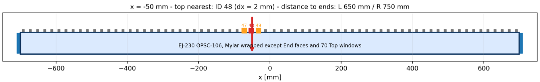


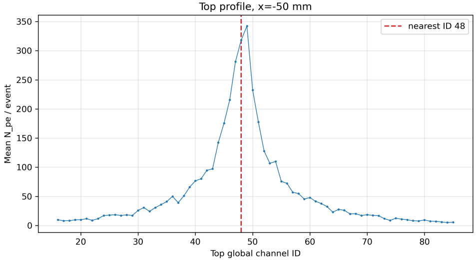
Quantity	Value
End L N_{pe}	203.9 ± 1.2
End R N_{pe}	115.5 ± 0.7
Nearest Top ID	46 (exact geometry)
Nearest Top N_{pe}	372.5 ± 2.2
Nearest Top $\langle t \rangle$	2.4913 ± 0.0019 ns
Nearest Top FPT	0.34565 ± 0.00016 ns

Run 14 | x = -100 mm | Photon arrival $\langle N(t) \rangle$



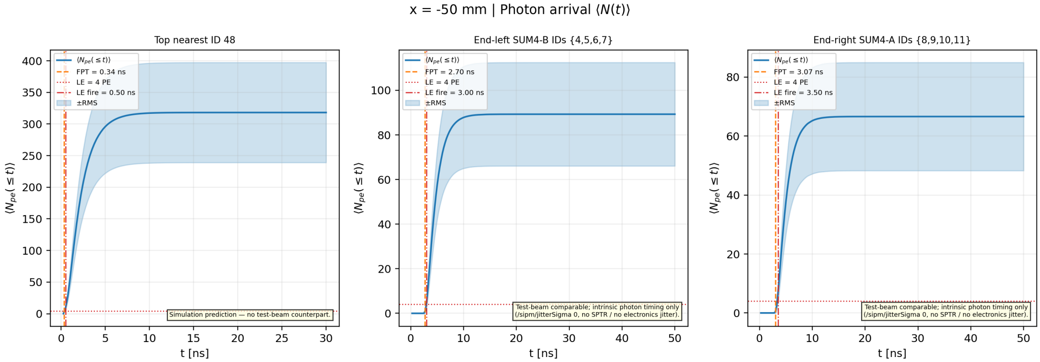
Position x=-50 mm



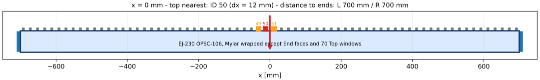


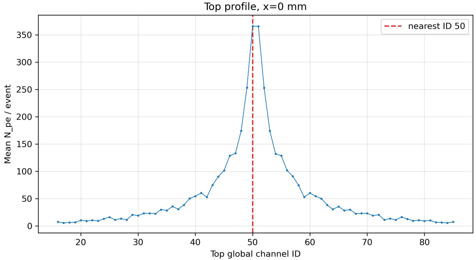
Quantity	Value
End L N_{pe}	176.1 ± 1.0
End R N_{pe}	131.0 ± 0.7
Nearest Top ID	48 (window-track exception)
Nearest Top N_{pe}	318.2 ± 1.8
Nearest Top $\langle t \rangle$	2.3422 ± 0.0020 ns
Nearest Top FPT	0.34234 ± 0.00017 ns

Run 15 | x = -50 mm | Photon arrival $\langle N(t) \rangle$



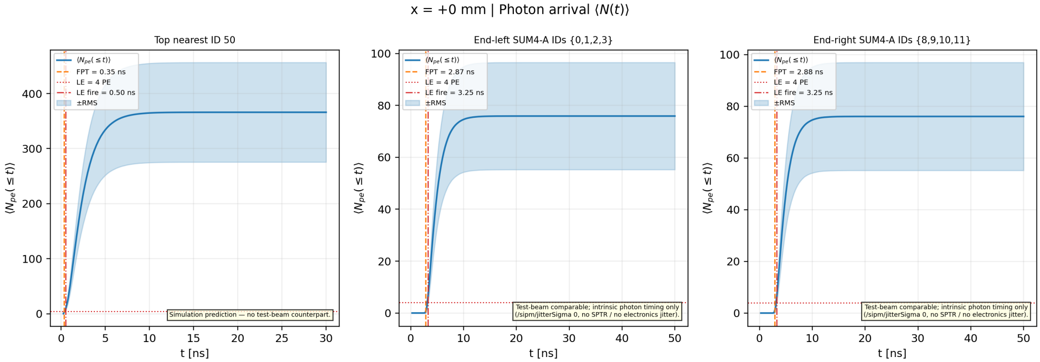
Position x=0 mm



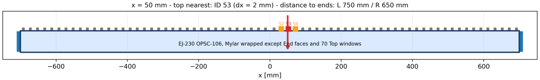


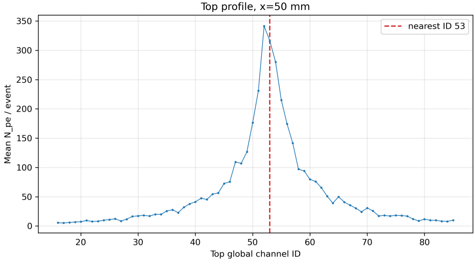
Quantity	Value
End L N_{pe}	151.0 ± 0.9
End R N_{pe}	151.1 ± 0.9
Nearest Top ID	50 (exact geometry)
Nearest Top N_{pe}	366.0 ± 2.0
Nearest Top $\langle t \rangle$	2.5528 ± 0.0020 ns
Nearest Top FPT	0.35099 ± 0.00023 ns

Run 16 | x = +0 mm | Photon arrival $\langle N(t) \rangle$



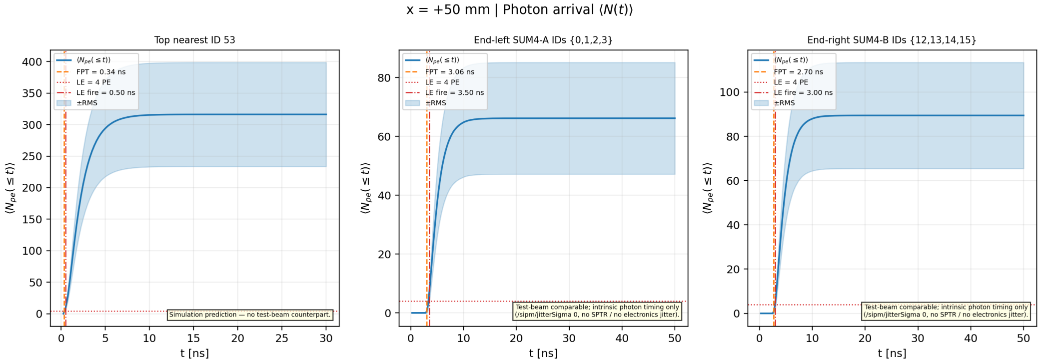
Position x=50 mm



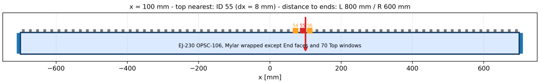


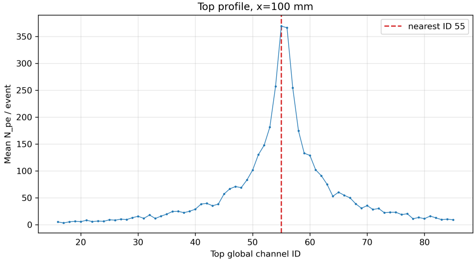
Quantity	Value
End L N_{pe}	129.9 ± 0.8
End R N_{pe}	175.9 ± 1.0
Nearest Top ID	53 (window-track exception)
Nearest Top N_{pe}	316.2 ± 1.8
Nearest Top $\langle t \rangle$	2.3413 ± 0.0020 ns
Nearest Top FPT	0.34218 ± 0.00014 ns

Run 17 | x = +50 mm | Photon arrival $\langle N(t) \rangle$



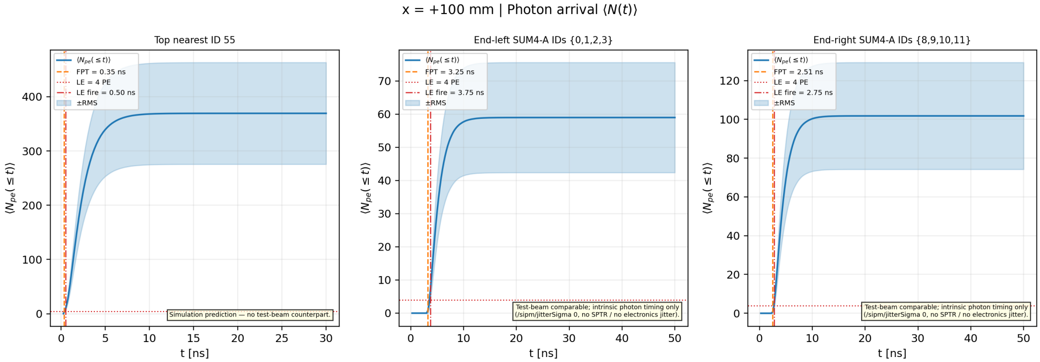
Position x=100 mm



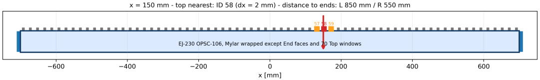


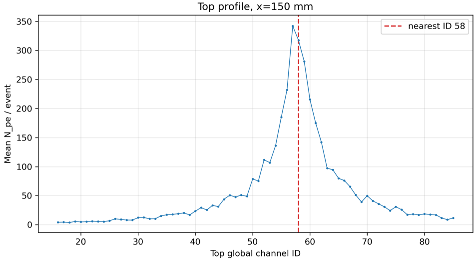
Quantity	Value
End L N_{pe}	114.4 ± 0.7
End R N_{pe}	202.3 ± 1.2
Nearest Top ID	55 (exact geometry)
Nearest Top N_{pe}	369.3 ± 2.1
Nearest Top $\langle t \rangle$	2.4904 ± 0.0019 ns
Nearest Top FPT	0.34584 ± 0.00017 ns

Run 18 | x = +100 mm | Photon arrival $\langle N(t) \rangle$



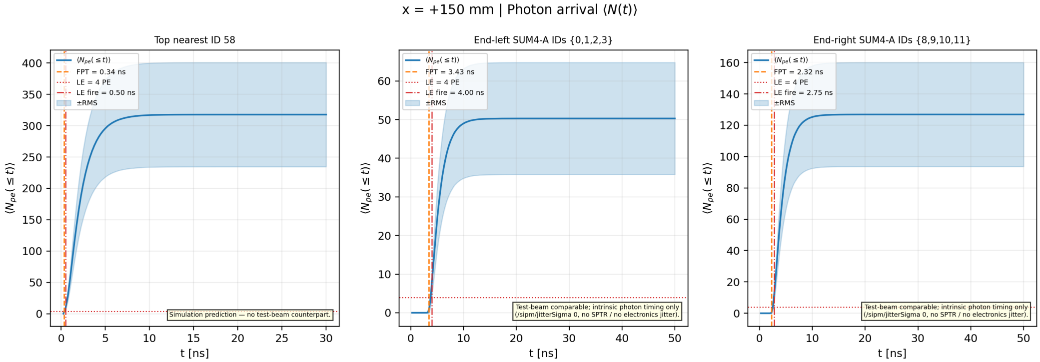
Position x=150 mm



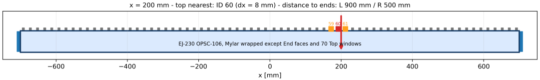


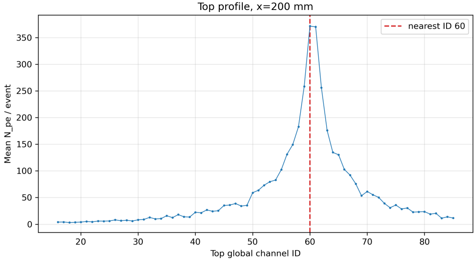
Quantity	Value
End L N_{pe}	100.5 ± 0.6
End R N_{pe}	240.4 ± 1.4
Nearest Top ID	58 (window-track exception)
Nearest Top N_{pe}	317.7 ± 1.9
Nearest Top $\langle t \rangle$	2.3405 ± 0.0020 ns
Nearest Top FPT	0.34244 ± 0.00018 ns

Run 19 | x = +150 mm | Photon arrival $\langle N(t) \rangle$



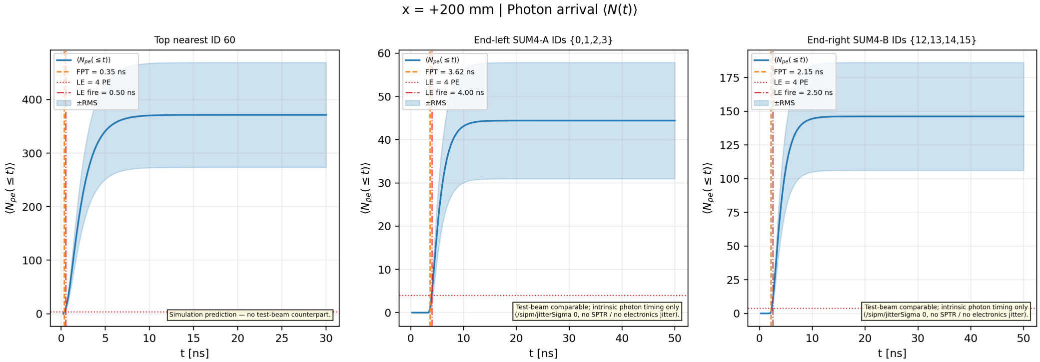
Position x=200 mm



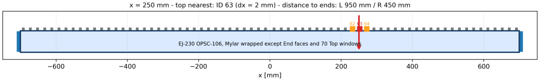


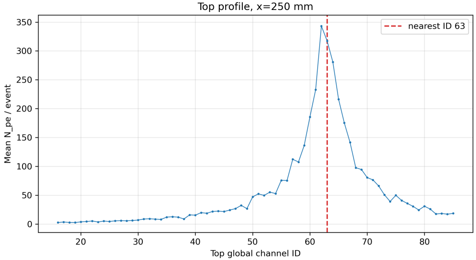
Quantity	Value
End L N_{pe}	88.0 ± 0.6
End R N_{pe}	285.6 ± 1.7
Nearest Top ID	60 (exact geometry)
Nearest Top N_{pe}	371.5 ± 2.2
Nearest Top $\langle t \rangle$	2.4939 ± 0.0019 ns
Nearest Top FPT	0.34568 ± 0.00021 ns

Run 20 | x = +200 mm | Photon arrival $\langle N(t) \rangle$



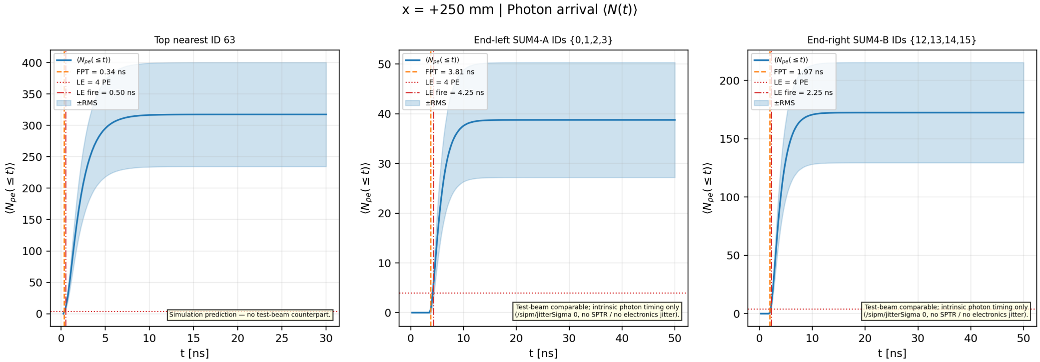
Position x=250 mm



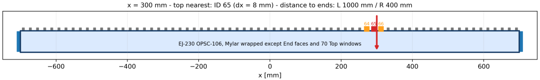


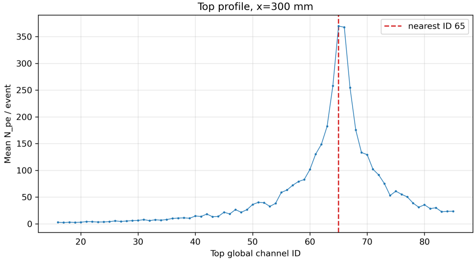
Quantity	Value
End L N_{pe}	76.7 ± 0.5
End R N_{pe}	337.3 ± 1.9
Nearest Top ID	63 (window-track exception)
Nearest Top N_{pe}	317.5 ± 1.9
Nearest Top $\langle t \rangle$	2.3414 ± 0.0020 ns
Nearest Top FPT	0.34205 ± 0.00013 ns

Run 21 | x = +250 mm | Photon arrival $\langle N(t) \rangle$



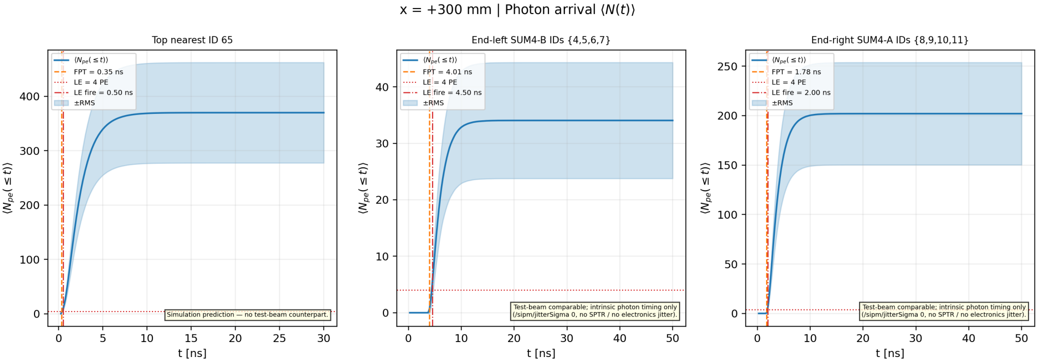
Position x=300 mm



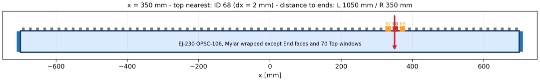


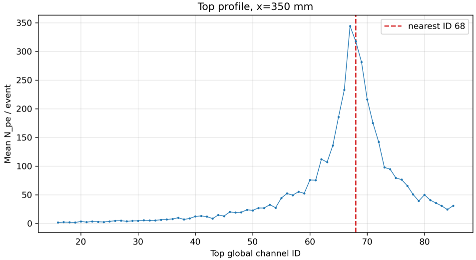
Quantity	Value
End L N_{pe}	67.4 ± 0.4
End R N_{pe}	396.4 ± 2.2
Nearest Top ID	65 (exact geometry)
Nearest Top N_{pe}	369.9 ± 2.1
Nearest Top $\langle t \rangle$	2.4957 ± 0.0020 ns
Nearest Top FPT	0.34550 ± 0.00014 ns

Run 22 | x = +300 mm | Photon arrival $\langle N(t) \rangle$



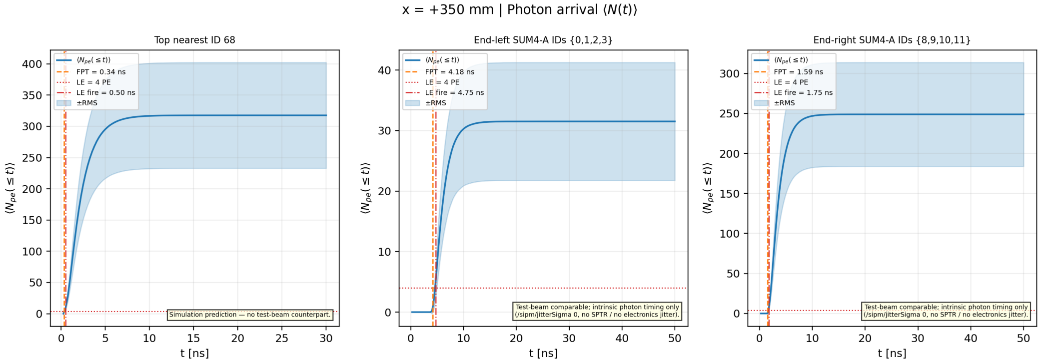
Position x=350 mm



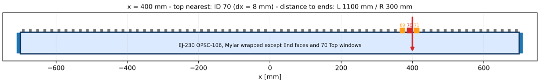


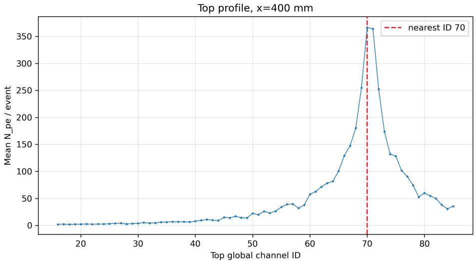
Quantity	Value
End L N_{pe}	60.6 ± 0.4
End R N_{pe}	480.0 ± 2.8
Nearest Top ID	68 (window-track exception)
Nearest Top N_{pe}	317.6 ± 1.9
Nearest Top $\langle t \rangle$	2.3425 ± 0.0020 ns
Nearest Top FPT	0.34251 ± 0.00021 ns

Run 23 | x = +350 mm | Photon arrival $\langle N(t) \rangle$



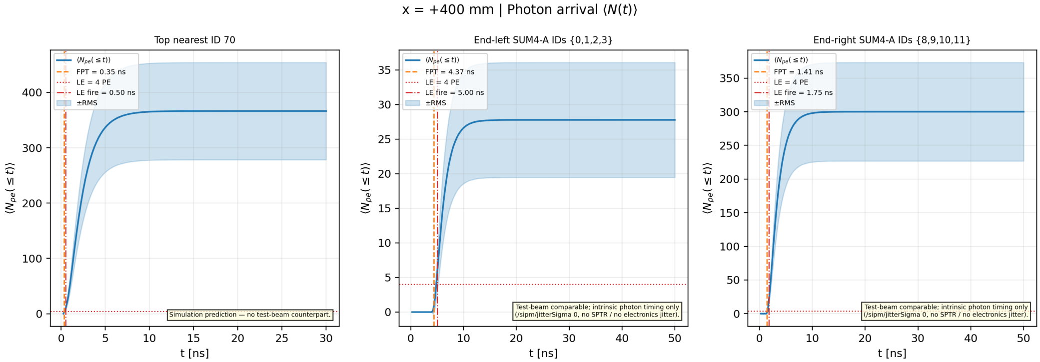
Position x=400 mm



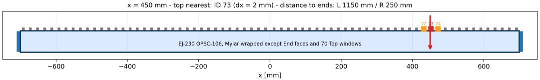


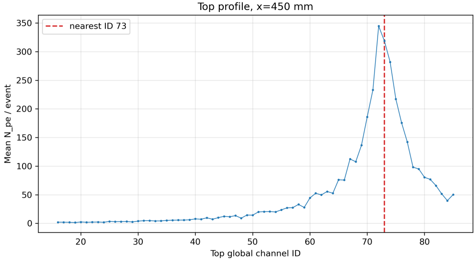
Quantity	Value
End L N_{pe}	52.8 ± 0.3
End R N_{pe}	577.8 ± 3.1
Nearest Top ID	70 (exact geometry)
Nearest Top N_{pe}	366.3 ± 2.0
Nearest Top $\langle t \rangle$	2.4909 ± 0.0020 ns
Nearest Top FPT	0.34615 ± 0.00025 ns

Run 24 | x = +400 mm | Photon arrival $\langle N(t) \rangle$



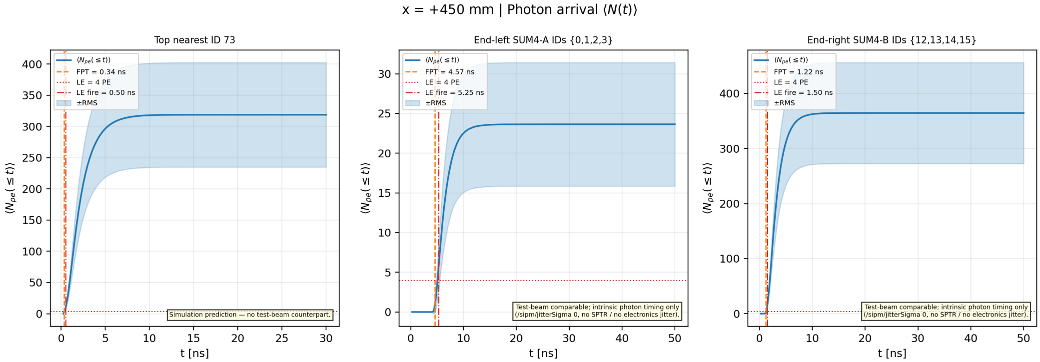
Position x=450 mm



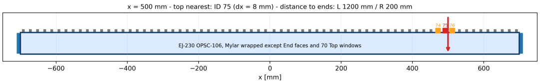


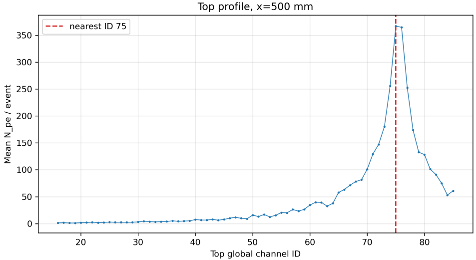
Quantity	Value
End L N_{pe}	47.0 ± 0.3
End R N_{pe}	719.4 ± 4.1
Nearest Top ID	73 (window-track exception)
Nearest Top N_{pe}	318.7 ± 1.9
Nearest Top $\langle t \rangle$	2.3429 ± 0.0020 ns
Nearest Top FPT	0.34232 ± 0.00013 ns

Run 25 | x = +450 mm | Photon arrival $\langle N(t) \rangle$



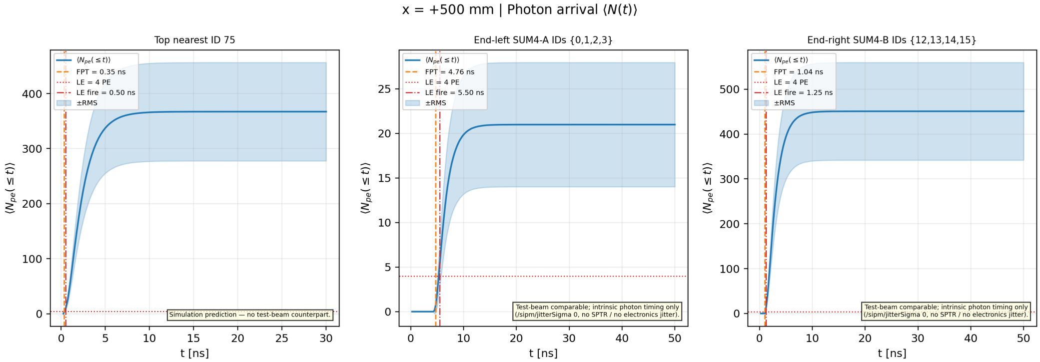
Position x=500 mm



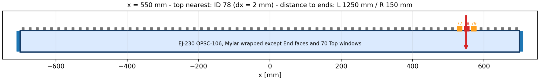


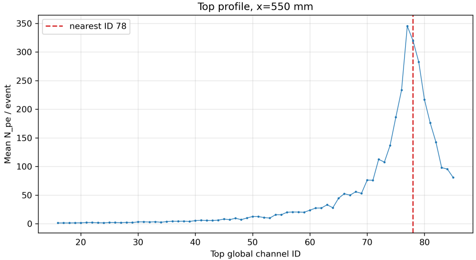
Quantity	Value
End L N_{pe}	41.6 ± 0.3
End R N_{pe}	890.1 ± 4.8
Nearest Top ID	75 (exact geometry)
Nearest Top N_{pe}	367.1 ± 2.0
Nearest Top $\langle t \rangle$	2.4910 ± 0.0020 ns
Nearest Top FPT	0.34605 ± 0.00020 ns

Run 26 | x = +500 mm | Photon arrival $\langle N(t) \rangle$



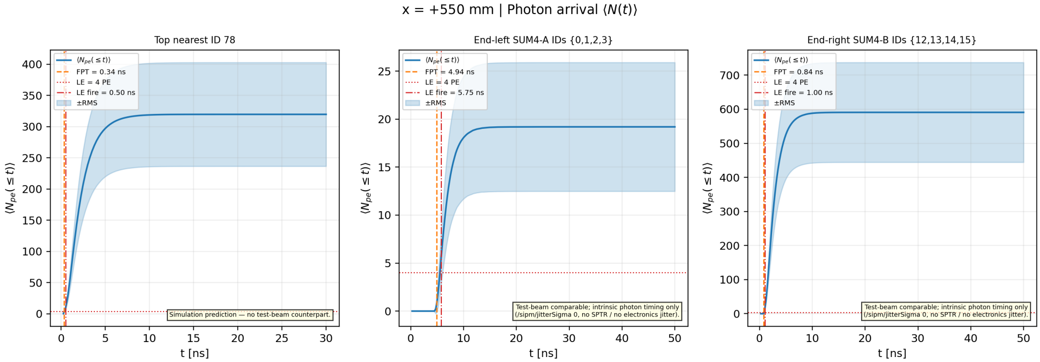
Position x=550 mm



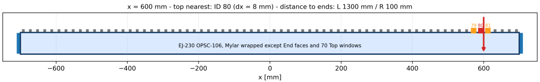


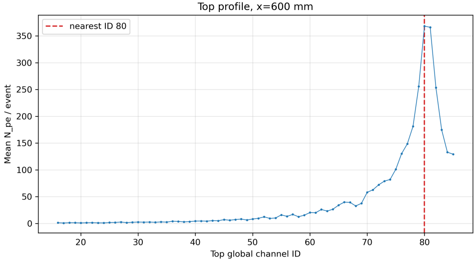
Quantity	Value
End L N_{pe}	37.8 ± 0.3
End R N_{pe}	1154.5 ± 6.5
Nearest Top ID	78 (window-track exception)
Nearest Top N_{pe}	319.6 ± 1.9
Nearest Top $\langle t \rangle$	2.3459 ± 0.0020 ns
Nearest Top FPT	0.34241 ± 0.00014 ns

Run 27 | x = +550 mm | Photon arrival $\langle N(t) \rangle$



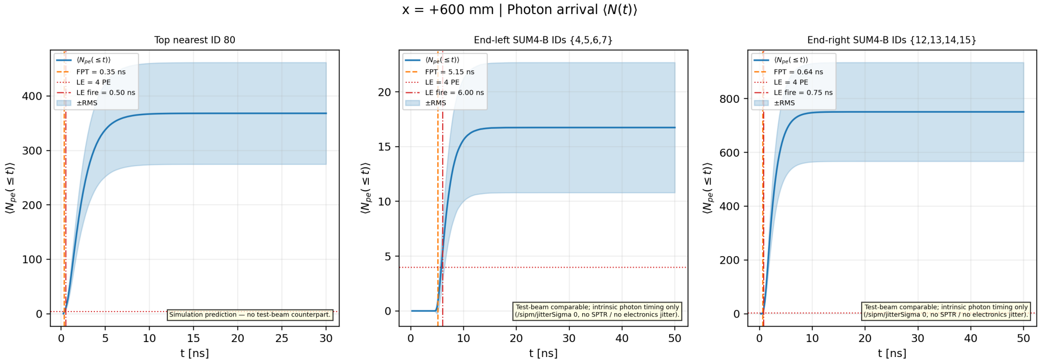
Position x=600 mm



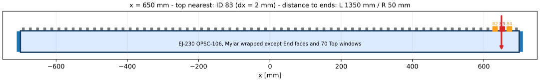


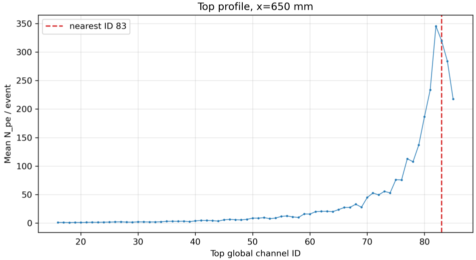
Quantity	Value
End L N_{pe}	33.1 ± 0.2
End R N_{pe}	1486.5 ± 8.1
Nearest Top ID	80 (exact geometry)
Nearest Top N_{pe}	368.2 ± 2.1
Nearest Top $\langle t \rangle$	2.4953 ± 0.0020 ns
Nearest Top FPT	0.34571 ± 0.00016 ns

Run 28 | x = +600 mm | Photon arrival $\langle N(t) \rangle$



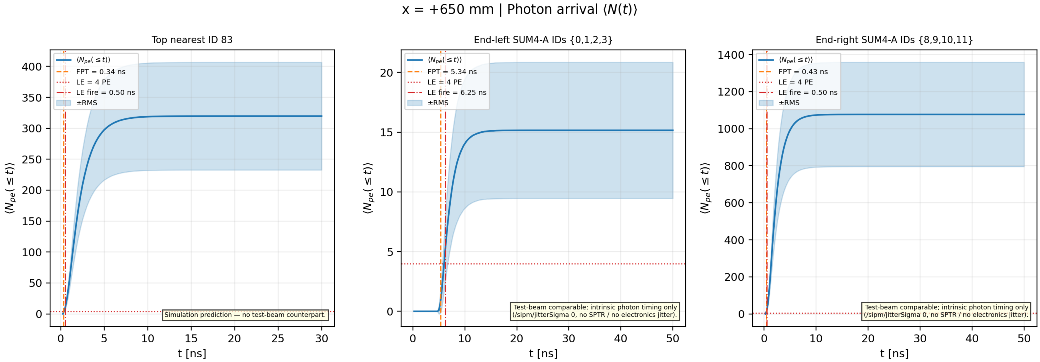
Position x=650 mm



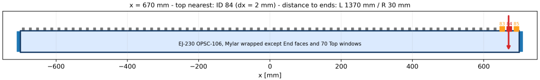


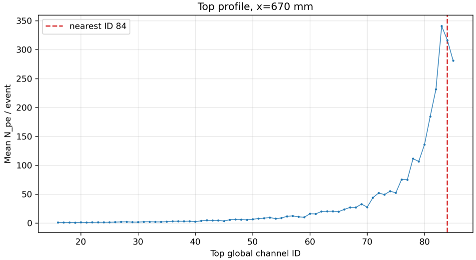
Quantity	Value
End L N_{pe}	30.2 ± 0.2
End R N_{pe}	2133.3 ± 12.4
Nearest Top ID	83 (window-track exception)
Nearest Top N_{pe}	319.7 ± 1.9
Nearest Top $\langle t \rangle$	2.3425 ± 0.0020 ns
Nearest Top FPT	0.34249 ± 0.00015 ns

Run 29 | x = +650 mm | Photon arrival $\langle N(t) \rangle$



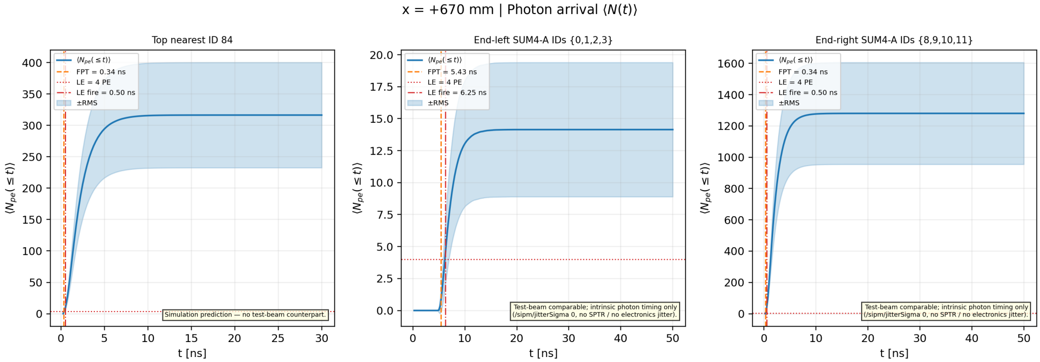
Position x=670 mm



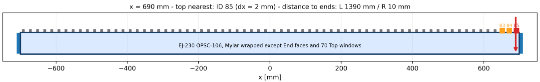


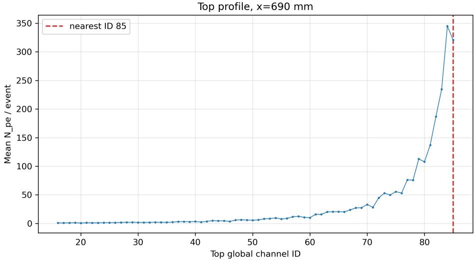
Quantity	Value
End L N_{pe}	28.2 ± 0.2
End R N_{pe}	2496.3 ± 14.1
Nearest Top ID	84 (window-track exception)
Nearest Top N_{pe}	316.4 ± 1.9
Nearest Top $\langle t \rangle$	2.3403 ± 0.0020 ns
Nearest Top FPT	0.34266 ± 0.00019 ns

Run 30 | x = +670 mm | Photon arrival $\langle N(t) \rangle$



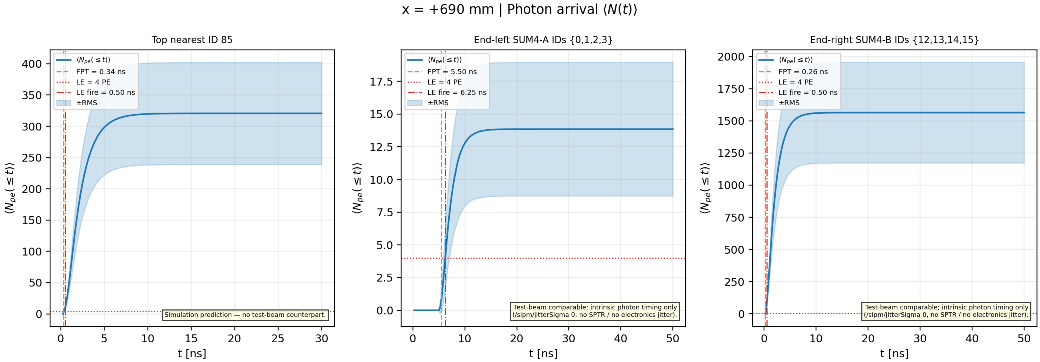
Position x=690 mm





Quantity	Value
End L N_{pe}	27.6 ± 0.2
End R N_{pe}	3089.6 ± 17.1
Nearest Top ID	85 (window-track exception)
Nearest Top N_{pe}	320.6 ± 1.8
Nearest Top $\langle t \rangle$	2.3455 ± 0.0020 ns
Nearest Top FPT	0.34224 ± 0.00015 ns

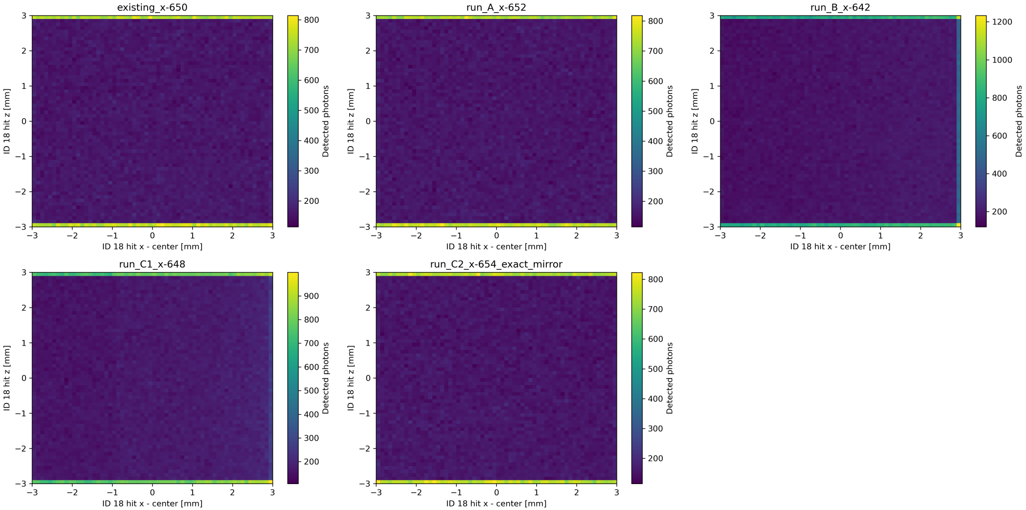
Run 31 | x = +690 mm | Photon arrival $\langle N(t) \rangle$



Backup: validation and gates

- 31/31 main ROOT files pass: event IDs 0–1999, matching beam x, aggregate IDs 0–85.
- Top localization passes all positions under the documented window-track/statistical-tie rules.
- At x=300 mm the two candidates differ by 0.00 sigma.
- MPT checks: RINDEX 1.58, yield 9700/MeV, decay 1.5 ns, rise 0.5 ns, ABSLENGTH 120 cm, emission peak $\approx$ 391 nm.

Backup: ID18 impact maps



Backup: why a 4-PE leading-edge threshold

1. **Noise rejection at hardware parity:** 4 PE is the nominal fixed threshold outside the correlated-noise spectrum.
2. **Slope optimum:** lower thresholds expose first-photon order statistics; higher thresholds reduce slope and increase walk.
3. **Far-end efficiency:** Minimum signal in the scan is End R 27.4 ± 0.2 at $x = -690$ mm; the nominal 4 PE threshold remains far below that signal.
4. **Walk:** no time-walk correction is applied in this campaign.

Backup: timing methodology

- SUM4 groups: $\{0..3\}$, $\{4..7\}$, $\{8..11\}$, $\{12..15\}$.
- Single-PE response: normalized 0.5/5 ns double exponential.
- Absolute leading-edge threshold: nominal 4 PE equivalent; the CSV-derived far-end rationale is reported on the preceding backup slide.
- Estimator: $\sigma(\Delta T_{AB})/\sqrt{2}$.
- No time-walk correction is applied in this campaign.
- The EndTop/End-only anti-artifact audit uses a common threshold, pulse, primary time convention and estimator.

Backup: glossary

FPT First photon time: first detected photon per event.

Fano factor $F \equiv \mathrm{Var}(N_{pe})/\langle N_{pe}\rangle = 1 + c\langle N_{pe}\rangle$ when Landau ΔE fluctuations are present.

Group A / Group B Same-End SUM4 groups: A={0–3}, B={4–7} on End L; mirrors {8–11}, {12–15} on End R. $\sigma_{\mathrm{group}} = \sigma(\Delta T_{AB})/\sqrt{2}$ (`sum4_timing.C:218-221`).

Time-walk No time-walk correction is applied in the EJ-230 campaign.

Intrinsic sigma Excludes SPTR and electronics jitter.

Moyal Closed-form Landau approximation for stable MINUIT fits. Undershoots extreme tail.

SPTR Single Photon Time Resolution: SiPM intrinsic single-PE jitter. Excluded from all EJ-230 results; all reported timing is intrinsic.

SUM4 Analog sum of four SiPMs: {0–3}, {4–7}, {8–11}, {12–15} (`congruent_sum4_timing.C:218-221`).

Timing gate `exec08b_timing_gate.py` tests EndTop vs. End-only σ_{group} at $x = 0, \pm 400$ mm; no time acceptance cut on photons.

Window-track effect Local symmetric ± 2 mm effect: 7.9% with significance 9.3–9.7σ; centered/midpoint contrast significances: -1.02σ/-0.06σ.

t_N / time-to-threshold N-th detected photoelectron arrival time per event. $\sigma(t_N)$: photon-counting timing resolution at ideal N-PE digital threshold.

(exact geometry) Nearest SiPM = maximum, or window-track exception; $x = 300$ mm is a statistical tie.

Backup: EndTop versus End-only tail audit